



Logistics Services and Transportation Sector Skills Framework

Sector Overview, Job Families and Pathways



**Issued by Logistics Services and Transportation Sector
Skills Council**

2025

Content disclaimer

This report is prepared **by the Logistics Services and Transportation Sector Skills Council**, Kingdom of Saudi Arabia, in 2025. It is **intended for general informational purposes** only. The information present in this document is **compiled from sources considered reliable and accurate** by the Logistics Services and Transportation Sector Skills Council and its participants. Although the Logistics Services and Transportation Sector Skills Council exerts the utmost effort to ensure **the accuracy and comprehensiveness** of the information provided in this report.

However, the Logistics Services and Transportation Sector Skills Council, its members, contributors, and the Ministry of Human Resources and Social Development **are not liable for any direct, indirect, or consequential loss or damage that may arise from the use of the information** contained in this report. The contents of this document report remain the property of the Logistics Services and Transportation Sector Skills Council.

Content

	Foreword	04
	Objectives of Logistics Services and Transportation Sector Skills Council and key contributors	05
01	Introduction	06
	1.1 Sector Skills Frameworks and their relevance to different stakeholders	07
	1.2 Development approach and methodology	08
02	Sector Skills Frameworks' applications	09
03	Logistics Services and Transportation Sector Skills Framework overview	11
	3.1 Key sector highlights and sub-sectors' overview	12
	3.2 Sector's predominant transformations and their impact on skills needs	16
04	Sector's value chains, job families and their pathways	27
	4.1 Logistics Services and Transportation value chain	31
	4.2 Travel value chain	33
	4.3 Job family profiles and pathway maps	35
05	Appendix	95
	5.1 Skills needs emerging from sector transformations	96
	5.2 Sources	104
	5.3 Glossary	105
	5.4 Abbreviations	107
	5.5 Acknowledgements	108

Foreword

Based on the Council of Ministers Resolution #254 dated 23/04/1442, the Labor Market Strategy was approved, aiming to initiate fundamental reforms in the labor market through a series of targeted initiatives. In response, the Ministry of Human Resources and Social Development, through its Skills and Training deputyship, has activated the Occupational Planning and Standards initiative, specifically via the key Sector Skills Councils project.

The Sector Skills Councils, composed of representatives from sector employers, line ministries, and other key stakeholders, have been instrumental in developing comprehensive Sector Skills Frameworks. These frameworks are carefully designed to identify the essential skills required across various sectors, ensuring that our workforce is well-prepared to thrive in a competitive global marketplace.

We are proud to present the Sector Skills Framework for the Logistics Services and Transportation Sector. This framework is the culmination of rigorous research and extensive stakeholder engagement. It delineates the technical and general skills required within the sector and provides a formalized articulation of the labor market's skills needs. Aligning skills development with the skills needed by the labour market, the framework aims to enhance employability, foster economic growth, and contribute to the sustainable development of the Kingdom.

We extend our sincere gratitude to all members of the Sector Skills Council, sector employers, educational institutions, and other stakeholders who have contributed to the development of this framework. The successful implementation of this framework will undoubtedly lead to a more competent and competitive workforce, driving innovation and productivity within the Logistics Services and Transportation Sector. Together, we can ensure that the Kingdom of Saudi Arabia remains at the forefront of global economic progress.



Objectives of Logistics Services and Transportation Sector Skills Council

The Logistics Services and Transportation Sector Skills Council aims to achieve the following objectives:



- Determine requirements, standards and career pathways to support planning for professions and skills
- Establish consolidated sector intelligence with implications on current and future skills needs and challenges
- Align training and education programs with labor market needs
- Support in bridging skills gaps and skills shortages, and improving labor productivity
- Improve workforce skills across the sector

Key contributors

Sector Skills Council Members



Badr Alghazi
(SSC Chairperson)

Advisor on Human Capability Development



“The human capital element is a key the pillar of the National Strategy for Logistics Services and Transportation Sector, hence developing this element and raising and refining its skills is one of the priorities that we are keen on. This project came at the right time to integrate the sector's efforts in focusing on raising the skills of our talent. Identifying and developing the basic skills needed by the main economic activities is one of the important factors for achieving sustainable development by empowering and supporting human capabilities to achieve national targets.”



Dr. Omairah Bamasag
DM of Transportation Enabling



Sector Expert



Saeed Alzahrani

Senior Director, Talent Acquisition and Organization Development



Waleed Almalki

VP of Human Resources



Sultan Alghamdi

Shipa by Agility General Manager



Mohammed Fallatah

General Manager, Human Capital



Naif Alzahrani

Senior HR Business Partner



Rakan Alotaishan

Representative of Federation of Saudi Chamber for Logistics Services and Transportation SSC



Khalid Albusayyes

PIF Representative for Logistics and Transportation SSC



Hisham AlKhalidi

Chief Sector Officer



Eng. Abdulrahman Turkistani

GM of HR and Business Support



Samir Berqdar

Chief Executive Office



Deena Faki

Area HR Manager



Fawaz Almuhanaa

Regional HR Manager



Dr. Masha'al Alqahtani

Director of Empowerment and Nationalization



Sultan Al-Rasheedi

Executive Director of Business Development



01

Introduction

This report is built to **detail out different components of the Sector Skills Framework for the Logistics Services and Transportation Sector in Saudi Arabia**. The report is to be used in tandem with **Logistics Services and Transportation Skills Dictionary Report** and the **Logistics Services and Transportation Skills Repository Excel Tool**.

The primary purpose of this report is to **outline the sector overview, its sub-sectors, key sector transformations and skills needs in line with sector transformations, as well as define job families and career pathways in the Logistics Services and Transportation sector**. Clearly established job families and career pathways will equip the sector stakeholders to make informed decisions with regards to skilling efforts, trainings required, sector policymaking, performance improvement, etc.

User can access Skills Dictionary Report

by clicking below



User can access Skills Repository Excel Tool

by clicking below



Sector Skills Frameworks and their relevance to different stakeholders

A Sector Skills Framework (SSF) is a comprehensive tool that **outlines the skills needs within the sector**. It identifies the **skills required for each job family** across the sector's value chain(s). The Sector Skills Framework follows a **structured approach** to identify the skills required within the sector wherein first job families across sub-sectors are identified, followed by outlining key job roles, pathways and priority skills required.

The following is how the Sector Skills Frameworks are intended to support **key stakeholders of the Logistics Services and Transportation sector**:



Employed and unemployed workforce

Workforce in the sector can **clearly understand the skills required** and **plan their professional development**. This will also facilitate mobility across jobs and enable unemployed members to **prepare for job roles of interest**.



Employers

Employers can use the Sector Skills Framework to **plan employee-trainings in line with industry standards** for skills and **support skills-based hiring** rather than occupations-based hiring.



Education and training entities

Education and training entities can use the framework to **assess alignment between learning outcomes of programs and skills identified in the Sector Skills Frameworks**. They can also leverage the framework to design programs based on current and future industry needs of the sector.



Policymakers

Policymakers can use Sector Skills Frameworks to **understand industry's skills needs** and tailor policies to **address skills needs and shortages**.



Funding entities

Funding entities can refer to Sector Skills Frameworks as **a guiding tool to identify areas that require funding** and enable optimized allocation of funds in the sector.



Accreditation and licensing entities

Accreditation and licensing entities can use the Sector Skills Framework to **enhance accreditation and licensing process** in line with industry priority skills needs.



Rakan Alotaishan

Representative of Federation of Saudi Chamber for Logistics Services and Transportation SSC



“Development of the Sector Skills Framework will enable decision-makers to identify the skills required in the sectors. These frameworks are particularly effective as they are shaped by practitioners and experienced professionals from diverse backgrounds within the sector.”

02

Sector Skills Frameworks' applications

Sector Skills Frameworks are a critical tool primarily for two key sector stakeholders: **(1) Employers (2) Education and training entities**. The frameworks act as a guide **for these stakeholders to develop policies, curricula, skills development initiatives and organizational policies**. As part of this report, **multiple specific use-cases** of Sector Skills Frameworks are identified for these stakeholders and are present in the subsequent section.

Sector Skills Frameworks use-cases for employers and education and training ecosystem entities

Employers



Sector Skills Framework is an important tool for employers to enable targeted workforce development and recruitment. It helps align employers' workforce capabilities/ requirements with evolving industry needs and standards. The framework facilitates several applications:

- **Market segmentation:** Adopting SSC sector/ sub-sector definitions to unify language across the skills ecosystem to facilitate cross-entity and cross-sectoral discussions, engagements, and initiatives, for example, implementing unified language and definitions across Air Transport, Land Transport and Maritime Transport sub-sectors
- **Career pathway mapping:** Providing employers with career pathway maps at the job family level, helping to define progression plans, for example, identifying typical roles and their transitions within Engineering and Technical Service job family
- **Skills gaps identification:** Conducting skills gap analysis with Sector Skills Framework as the base, in order to ensure that the entity remains competitive by aligning its skills with evolving demands, for example, reviewing Sector Skills Framework and identifying growing demand for digitalization in the sector
- **Unified job family classification:** Categorizing new and existing efforts under the relevant job family by sub-sector, for example unifying definition of the "E-logistics" job family
- **Job roles definition:** Defining job roles and responsibilities in line with identified priority skills across different job families facilitating targeted hiring and setting job expectations, for example, including key skills requirements for an Airport operations manager in the job definition

Education & training entities



Sector Skills Framework is a reference point for education and training entities, helping them ensure their programs target priority skills in alignment with industry skills needs. There are multiple facets through which Sector Skills Framework enables this motive:

- **Funding prioritization:** Prioritizing funding programs aligned with high-priority skills highlighted within Sector Skills Framework, for example, partnering with academies offering warehouse management trainings
- **Program and curricula design:** Ensuring current and new programs address skills needs based on the Sector Skills Framework, for example, comparing skills covered in current programs versus identified skills needs for a refined curricula.
- **Training gaps identification:** Ensuring coverage of education and training offerings across the job families identified by the Sector Skills Framework and targeting the right skills needed through a program, for example, launching specialized training modules for supply chain management for particular levels
- **Licensing and accreditation:** Facilitating licensing and accreditation of programs addressing priority skills needs as identified by the Sector Skills Framework, for example facilitating the accreditation process for autonomous transportation skills

03

Logistics Services and Transportation Sector Skills Framework Overview

Logistics Services and Transportation Sector Skills Framework is built leveraging the following sector definition for **the Logistics Services and Transportation Sector**:

Logistics Services and Transportation includes provision of **passenger or freight transport** by rail, road, maritime or air and associated activities. It covers **logistics services, supply chain services, operation and maintenance of facilities** such as terminal and parking facilities, cargo handling, storage, etc.

The subsequent section provides an overview of the Logistics Services and Transportation sector highlights and definitions for the three Logistics Services and Transportation sub-sectors:
(1) Land Transport (2) Maritime Transport (3) Air Transport

Key sector highlights

~197
Billion

Size of Logistics Services and Transportation GDP in KSA in SAR (2022)¹

6%

Contribution % of Logistics Services and Transportation sector to GDP in KSA (2022)



The Kingdom is home to **211** road and public transport lines, **3** rail lines, **10** ports, and **27** airports; with **91% of the population within a 120-minute drive** of an airport (ranked 1st globally for road connectivity)



Logistics Services and Transportation represents **~6% of overall KSA GDP** (SAR ~197B as of 2022), aiming to reach **~10%** by 2030



The Logistics Services and Transportation private sector employs **~359K of KSA's labor force** of which **30%** (~107.7K) are Saudis and **9%** (31.5K) are female²



The Kingdom's Freight and Logistics **market size is estimated at ~95B SAR** in 2024, and is expected to reach ~123B by 2029; growing at a CAGR ~5%

Note: The SSF does not consider activities: pipeline transportation (classified under Energy and Utilities Sector Skills Framework), construction related assets such as ports, airports or rail (classified under Real estate and development); Sector maps to the Nat. classification for economic activities section H, divisions 49, 50, 51, 52, 53 and section N, division 77 group 771

Source: (1) World Economic Forum Global Competitiveness Report (2019); (2) MHRSD Business Intelligence (2023) data covering the private sector (linked to ISIC section H, divisions 49, 50, 51, and 52); National classification of economic activities

Logistics Services and Transportation sub-sectors' overview



Land Transport

Sub-sector level 1

- **Land transport and transport via pipelines:** Passenger and freight transport over land through rail and roads
- **Warehousing and support activities for transportation:** Operations and maintenance of passenger and freight transport over land through rail and road



Maritime Transport

Sub-sector level 1

- **Maritime transport:** Maritime transport of passengers and freight
- **Warehousing and support activities for transportation:** Shipping, port operations and maritime services



Air Transport

Sub-sector level 1

- **Air transport:** Transport of passengers or freight by air or space
- **Warehousing and support activities for transportation:** Airport Operations



Sara Aloraimi
SNC Representative for
Logistics Services and
Transportation SSC



The Logistics Services and Transportation SSF identifies the essential skills requirements, enhancing operational efficiency and workforce development. It contributes to improved supply chain management, increases customer satisfaction, and strengthens the competitiveness of companies.

Logistics Services and Transportation sub-sectors' overview

Details of the Logistics Services and Transportation sub-sectors

Land Transport

Sub-sector level 1	Definition	Sub-sector level 2	Sample companies
Land transport and transport via pipelines	Passenger and freight transport over land through rail and roads	Transport of passengers and freight via railways	  
		Other land transport (including Freight transport by road)	
Warehousing and support activities for transportation	Operations and maintenance of passenger and freight transport over land through rail and road	Support activities for transportation for passengers or freight (Service activities to incidental land transport, including operation and maintenance of railway facilities and bus stations)	   
		Warehousing and storage (operation of storage and warehouse facilities for all kind of goods)	

Maritime Transport

Sub-sector level 1	Definition	Sub-sector level 2	Sample companies
Maritime transport	Maritime transport of passengers and freight	Sea and coastal maritime transport (excluding Inland maritime Transport)	
Warehousing and support activities for transportation	Shipping, port operations and maritime services	Support activities for transportation (Service activities incidental to maritime transportation, including terminal facilities operations, ships operations, navigation)	
		Warehousing and storage (operation of storage and warehouse facilities for all kind of goods)	

Logistics Services and Transportation sub-sectors' overview

Details of the Logistics Services and Transportation sub-sectors

Air Transport

Sub-sector level 1	Definition	Sub-sector level 2	Sample companies
Air transport	Transport of passengers or freight by air or space	Passenger air transport	
		Freight air transport	
Warehousing and support activities for transportation	Airport Operations	Support activities for transportation (Service activities incidental to air transportation incl. airport activities and navigation, and cargo handling)	 
		Warehousing and storage (operation of storage and warehouse facilities for all kind of goods)	



Mohammed Fallatah

General Manager, Human Capital



“ The Sectoral Skills Framework involves analyzing the sector's nature, simplifying its complexities, and presenting clear insights to stakeholders. This framework aims to enhance productivity, ensure continuity, and foster development and sustainability.



Sector's predominant transformations and their impact on skills needs

01

Global transformations

As part of the global transformations, major **developments, transitions, and patterns** are being observed around the world within the Logistics Services and Transportation sector, such as **Technology and digitalization, Environmental, social and governance, Customer preferences and Business ecosystem**.

02

Relevant local sector transformations and strategic imperatives

With respect to local sector transformations and strategic imperatives, there are **significant advancements and shifts occurring specifically within KSA**. These include initiatives aligned with **National priorities**, comprehensive **National strategies** (such as **National Industrial Development and Logistics Program (NIDLP), National Transports and Logistics Strategy, Saudi Aviation Strategy**, etc.), and substantial **investments**. Additionally, **private and corporate initiatives** are playing a key role in shaping industry's landscape.

03

Workforce transformations

In relation to workforce transformations, noticeable changes, patterns, and shifts are identifiable in factors like **High demand for labor, Demand for specialized skillset, Increased focus on labor safety, and Limited workforce**.

01 Global transformations

The Logistics Services and Transportation sector is undergoing rapid transformations globally, shaped by numerous transformations across four main categories: **(1) Technology and digitalization (2) Environmental, social and governance (3) Business ecosystem, and (4) Customer preferences**

Technology and digitalization

Within Technology and digitalization, **8 main transformations** in the following spaces - Internet of things (IoT), Analytics and artificial intelligence, Blockchain, Autonomous transportation, Advanced autonomous tools, High speed rails, Single portal, and 3D printing technology, **are collectively changing the Logistics Services and Transportation industry and workforce.**



Internet of things (IoT)

The integration of IoT devices are **transforming fleet and logistics management by seamlessly connecting operations to monitor fleets, track goods, and forecast equipment failures.** By leveraging real-time data, the devices **optimize routes, streamline inventory management,** and enhances driver behavior monitoring.



Analytics and artificial intelligence

Advanced analytical systems in the sector are **transforming operations by analyzing vast amounts of data to generate actionable insights.** Third-party logistics providers leverage data-driven decision-making to optimize routes, manage inventory, and forecast demand, **driving efficiency and precision** in supply chain management.



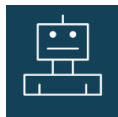
Blockchain

Enhanced security, transparency, and traceability through blockchain technology are revolutionizing supply chains by streamlining operations and reducing costs. For third-party logistics providers, this ensures **accurate inventory management,** swift dispute resolution, and reliable verification of freight documentation.



Autonomous transportation

Advancements in self-driving vehicles, drones, and automated port systems **are revolutionizing the transportation of goods and infrastructure management.** Coupled with automation and electrification, these innovations incorporate machine learning, predictive maintenance, and smart ticketing systems to optimize mobility solutions.



Advanced autonomous tools

The integration of robotics is **revolutionizing supply chain operations by minimizing human error and enhancing efficiency.** Combined with automated systems like WMS and TMS for tasks such as truck loading and material handling, it **streamlines warehousing functions** and optimizes the entire supply chain process.



High speed rails

High-speed rail technologies, like Hyperloop, are gaining momentum due to their **ability to drastically reduce travel times, ensure exceptional safety and reliability,** and promote environmental sustainability, redefining the future of transportation.



Single portal

Single portal system is revolutionizing trade and transport **by enabling all involved parties to submit standardized information and documents through a single-entry point.** This streamlines compliance with import, export, and transit regulatory requirements, enhancing efficiency and reducing complexity in global trade operations.



3D printing technology

On-demand production powered by 3D printing technology is **transforming logistics by reducing inventory needs and optimizing warehousing space.** This innovation enables more flexible and responsive solutions, **allowing businesses to adapt quickly to changing demands** and streamline operations

01 Global transformations

Environmental, social and governance

Within **Environmental, social and governance**, **3 transformations** are identified - Environmental regulations and governance, Supply chain sustainability, and Circularity, that are collectively transforming the overall Logistic Services and Transportation sector.



Environmental regulations and governance

Regulatory measures are being implemented to **curb GHG emissions in Logistics Services and Transportation**, targeting sustainability across the sector. Organizations like the International Maritime Organization and European Union have introduced regulations addressing ocean freight's environmental impact, including carbon tax initiatives, to promote greener operations.



Supply chain sustainability

Sustainable technologies, such as green logistics, alternative fuels, and e-vehicles, are **reshaping the logistics landscape to minimize environmental impact**. Coupled with infrastructure advancements like charging stations and power outlets, these efforts **focus on reducing waste and resource leakage**, fostering a cleaner and more efficient supply chain.



Circularity

Reverse logistics is **driving circular supply chains** by enabling the collection, refurbishment, and redistribution of used products. This approach **reduces waste, maximizes resource utilization, and supports sustainable practices**, transforming the way businesses manage end-of-life product cycles.

Business ecosystem

Within Business ecosystem, **7 transformations** are identified in the following spaces: E-commerce, supply chain robustness, Partnerships, New business models, Transport hub privatization, Growth of cost-efficient carriers, and Nearshoring. These transformations are leading to a **shift towards resilience, collaboration, and adaptability** in response to evolving market demands.



E-commerce

The rapid growth of online shopping has fueled a surge in **demand for last-mile delivery services, prioritizing faster shipping times and innovative delivery methods**. This shift is reshaping logistics operations to meet consumer expectations, **driving efficiency and adaptability in the supply chain**.



Supply chain robustness

A strong focus on supply chain connectivity is **enhancing infrastructure utilization and boosting flexibility** in supply chain management. This **enables businesses to better navigate geopolitical uncertainties** and ongoing trade tensions, ensuring resilience and adaptability in global operations.



Partnerships

Strategic partnerships and collaboration with third-party logistics providers are **transforming supply chains by enabling integrated logistics solutions**. These alliances **enhance resilience, scalability, and efficiency**, empowering businesses to adapt to dynamic market demands and challenges seamlessly.

01 Global transformations



New business models

Innovative business models are emerging to **address evolving customer and market needs, driven by the growth of e-commerce**. Solutions like shared mobility, optimized last-mile logistics, and on-demand delivery, exemplified by Uber, DHL, and Jahez, are **reshaping how goods and services are delivered** with greater speed and efficiency.



Transport hub privatization

Governments are transitioning their role in infrastructure through the **privatization of airports, seaports, and train stations**. This shift opens **new funding opportunities**, enhances operational efficiency, and expands product offerings, paving the way for more **innovative and customer-focused transportation solutions**.



Growth of cost-efficient carriers

Growing consumer preference for leisure and cost-efficient carriers is **boosting non-aero revenue streams**. This shift drives efforts in cost optimization and operational efficiency, **aligning industry strategies with evolving customer demands**.



Nearshoring

Businesses are increasingly focusing on nearshoring suppliers, **sourcing goods and services from geographically closer locations**. This strategy **reduces supply chain risks, shortens lead times, and enhances overall efficiency**, ensuring more reliable and resilient operations.

Customer preferences

Within Customer preferences, **7 major transformations** are identified in the following spaces: Transparency and visibility, Personalization, Omnichannel utilization, Micro-mobility, Mobility as a service (MaaS), Private aviation services, and Multimodality transportation. Customer preferences bucket **reflects a shift toward seamless, customer-centric, and efficient logistics experiences**.



Transparency and visibility

The demand for real-time delivery visibility in B2C and pricing transparency in B2B is **reshaping logistics expectations**. Businesses are **prioritizing enhanced tracking capabilities and clear pricing structures** to meet customer needs, driving trust and efficiency across supply chains.



Personalization

Companies are adopting **flexible and personalized services to cater to individual preferences**, offering features like scheduled deliveries, tailored logistics solutions, and communication via preferred channels. With **custom subscription models, convenience services, and Mobility as a Service**, businesses are redefining customer-centric solutions for enhanced satisfaction.



Omnichannel utilization

Rising consumer expectations for omni-present product availability are driving the need for seamless alignment across all channels. This approach **minimizes risks from fluctuating demand patterns and supports high shipping order frequency**, ensuring reliability and customer satisfaction.

01 Global transformations



Micro-mobility

The use of **lightweight vehicles** such as electric scooters and bicycles is **rapidly increasing**, particularly in urban centers. These eco-friendly options provide **efficient, cost-effective solutions for short-distance travel**, aligning with sustainable urban mobility trends.



Mobility as a service (MaaS)

Digital platforms like Uber, Careem, and Lyft are transforming mobility by **integrating multiple transportation options into a single interface**. These platforms allow users to seamlessly plan and book their trips in one place, **enhancing convenience and streamlining urban travel**.



Private aviation services

The **demand for private aviation services is growing**, driven by the need for enhanced customer experiences in business and medical emergency transportation. With a focus on personalization and customer loyalty, these services are **redefining convenience and reliability in critical travel scenarios**.



Multimodality transportation

Multimodal travel platforms are **revolutionizing transportation by enabling passengers to book their entire journey across various modes**—trains, buses, and cars—in one place. This seamless integration **enhances convenience, streamlines planning, and creates a more efficient travel experience**.



Eng. Abdulrahman Turkistani

GM, Human Resources and Business Support



The Sector Skills Framework will serve as a cornerstone for the sector's sustainability, enhancing workforce readiness and ensuring alignment with the sector's ambitious aspirations and future opportunities..

02 Relevant local sector transformations and strategic imperatives

In KSA, **significant advancements and shifts** are happening within the local sector, driven by initiatives aligned with **National priorities and strategies**, and involvement of **private corporate** initiatives are shaping the industry

Saudi Arabia has launched multiple programs under **Vision 2030**, focusing on promoting the growth of the Logistics Services and Transportation sector, and improving the local, regional and international connectivity of trade and transport networks

The **National Industrial Development and Logistics Program (NIDLP)** helps achieve this goal by creating and improving the **performance of logistics hubs**. The program has added **5 new shipping lines** to the Kingdom's ports connecting Asia, Africa, and Europe, helped in signing the BOT agreement with the private sector in the Jeddah Islamic port to develop and operate container terminals with an investment of **~SAR 9 Bn**. The program has helped reorganize the **Saudi Arabia Railway Company** and the **Saudi Railway Organization** into one entity, resulting in performance improvement. It has also enhanced the airport infrastructure to enable the **growth of passengers**, reaching **~103M travelers** in **2019**

In addition to **Vision 2030** and its strategic program, there are national strategies that focus on development of the Logistics Services and Transportation sector. Major National strategies and initiatives for the sector includes:

Saudi Aviation Strategy focuses on **privatization of KSA airport** operations to increase the annual passenger traffic to **330M** and connectivity with **250+ destinations** across **29 airports**

National Transport and Logistics Strategy aims to **reduce energy consumption** and **transform** the kingdom into a logistics hub with the launch of a new national carrier and development of infrastructure (for e.g. expanding the rail network)

The PIF key sectoral initiatives (2021-2025) include investments in local and international companies to improve connectivity, infrastructure and operations

Saudia Group and the Ministry of Transport and Logistic Services sign memorandum of understanding to develop national capabilities aiming to **provide 553 scholarships focusing on academic and professional training in the aviation sector**

Complementing the efforts of the public sector, various development projects and investments are assisting the growth of the sector. Developments such as the **Riyadh metro, airports and airlines** such as Red Sea Airport and Riyadh Air have been implemented with projects like **Jeddah Islamic Port expansion** and the opening of **59 Logistics zones** as a part of sector's development plan

The **General Authority for Ports (Mawani)** launched the **Smart Ports initiative**, which aims to automate operations in the Kingdom's ports and it partnered with Maersk to develop a logistics park entirely powered by solar panels, covering over 65K sqm. In addition, advanced technology initiatives by private companies such as a new **logistics center** in Jeddah opened by **DHL Supply Chain** in 2019, **Noon** using **automated sorting systems** to streamline its warehousing and distribution operations, also stand out as key initiatives

03 Workforce transformations

In workforce transformations, there are noticeable shifts in factors such as **High demand for labor**, **Demand for specialized skillset**, **Increased focus on labor safety** and **Limited workforce**

High demand for labor

Due to COVID-19 and the surge in online shopping, **global e-commerce sales rose by 28% in 2020**, doubling the growth seen in previous years. This sales boom subsequently amplified the demand for deliveries, leading to increased labor requirements in the Logistics Services and Transportation sector. In response, the Kingdom is **actively promoting Saudization and talent development in the supply chain and logistics industries**. As part of these efforts, Saudi Arabia's General Transport Authority signed agreements with several private sector firms to localize as many as **13,000 jobs** for bus and truck drivers.



Demand for specialized skillset

The rising demand for innovative technologies, including **blockchain, robotics, and workplace automation**, is fueling a significant shift toward digitalization. Consequently, employers are placing greater emphasis on hiring talent with advanced technology-related skills to meet evolving industry needs. To support this transformation, initiatives **in collaboration with KAUST and Tabadul are being launched to encourage students** to engage in research and innovation projects and create solutions for advancement of global trade to KSA.

Increased focus on labor safety

The Logistics Services and Transportation sector is inherently high-risk, with workers frequently exposed to hazards such as heavy machinery, hazardous materials, and extended driving hours. In response, there is a growing global emphasis **on enhancing safety protocols and providing comprehensive training to reduce accidents and injuries**. In Saudi Arabia, initiatives such as the establishment of the National Road Safety Center prioritize workforce protection, focusing on public transportation personnel, service operators, and drivers. Furthermore, the General Transport Authority is enhancing safety regulations for maritime transport workers, addressing key areas like the safe manning of ships, the safety of life at sea, and compliance with executive regulations for safety management.

Limited workforce

The trucking sector within the transportation industry is facing a **significant driver shortage as many current drivers approach retirement age, and younger generations are not entering the profession at the same rate**. This challenge has prompted increased recruitment efforts, along with higher wages and enhanced benefits to attract new talent. While Saudi Arabia is not experiencing the demographic pressures of an aging population, it is proactively investing in education and training programs to prepare younger generations for future jobs. Initiatives such as the Saudi Logistics Academy and the Chartered Institute of Logistics Services and Transportation aim to equip individuals with the skills needed to support the evolving demands of the Logistics Services and Transportation sector.

Key skills needs in line with sector transformations



Technology and digitalization

Internet of Things (IoT)

- IoT applications
- Big data analytics
- Data governance
- Strategy/ processes design, project management, change management

Analytics and AI

- AI applications
- Data analytics
- Information technology and network security
- Strategy/ processes design, project management, change management

Blockchain

- Blockchain technology application
- Technology integration
- Information technology and network security
- Strategy/ processes design, project management, change management

Autonomous transportation

- Autonomous systems engineering and maintenance
- Data analysis and machine learning in transportation
- Automated logistics and port infrastructure management
- Strategy/ processes design, project management, change management

Advanced autonomous tools

- Automated warehousing systems integration
- Cobots in logistics
- Logistics risk and technology management
- Strategy/ processes design, project management, change management

High speed rails

- Passenger train operations
- Railway safety and risk assessment
- Rail regulatory compliance
- Strategy/ processes design, project management, change management

Single Portal

- Digital literacy
- System connectivity
- User experience design
- Strategy/ processes design, project management, change management

3D printing technology

- Rapid prototyping coordination
- 3D printing technical operation
- File preparation
- Strategy/ processes design, project management, change management

Key skills needs in line with sector transformations



Environmental, social and governance

Environmental regulations and governance

- Environmental protection management
- Corporate governance
- Risk and compliance reporting
- Strategy/ processes design, project management, change management

Supply chain sustainability

- Environmental sustainability management
- Energy efficiency
- Regulatory compliance
- Strategy/ processes design, project management, change management

Circularity

- Circular economy strategies
- Order fulfilment administration
- Regulatory compliance
- Strategy/ processes design, project management, change management

Key skills needs in line with sector transformations



Business ecosystem

E-commerce

- Order fulfilment administration
- Route planning and operations
- Customer management
- Strategy/ processes design, project management, change management

Supply chain robustness

- Demand forecasting
- Contingency planning
- Transportation route optimization
- Strategy/ processes design, project management, change management

Partnerships

- Third-party logistics provider selection and management
- Partnership management
- Supply chain partnership risk management
- Strategy/ processes design, project management, change management

New business models

- Last-mile delivery optimization
- On-demand logistics coordination
- Logistics risk management
- Strategy/ processes design, project management, change management

Transport hub privatization

- Transport hub investment analysis
- Regulatory knowledge
- Operational risk management
- Strategy/ processes design, project management, change management

Growth of cost-efficient carriers

- Logistics solutions sales
- Value engineering
- Transport cost optimization
- Strategy/ processes design, project management, change management

Nearshoring

- Cross docking
- Warehouse space utilization
- Warehouse inventory control/ audit
- Strategy/ processes design, project management, change management

Key skills needs in line with sector transformations



Customer preferences

Transparency and visibility

- Technology application
- Pricing for transportation services and operations
- Customer experience management
- Strategy/ processes design, project management, change management

Personalization

- Market knowledge
- User experience design
- Strategic service excellence
- Strategy/ processes design, project management, change management

Omnichannel utilization

- Analytical skills (incl. customer data analysis)
- Customer experience management
- Omnichannel supply continuity
- Strategy/ processes design, project management, change management

Micro-mobility

- Technology application
- Asset management
- Security management
- Strategy/ processes design, project management, change management

Mobility as a service (MaaS)

- Market research and analysis
- User experience design
- Cybersecurity
- Strategy/ processes design, project management, change management

Private aviation services

- Flight planning
- Flight dispatch
- Hazard and risk control and policy management
- Strategy/ processes design, project management, change management

Multimodality transportation

- Logistics product solution
- Hub operations
- Interchange operations
- Strategy/ processes design, project management, change management

04

Sector's value chains, job families and their pathways

To support workforce development in the Logistics Services and Transportation sector and align with the National ambitions, Logistics Services and Transportation Sector Skills Framework report **outlines the sector's job families and career pathways**. By **mapping out sub-sectors, defining value chains and detailing job families across the value chains**, this report offers a **structured view of the opportunities and skills requirements** across the Logistics Services and Transportation sector. The purpose of this section is to **serve as a resource** for anyone interested in understanding the **diverse roles within the Logistics Services and Transportation sector, identifying key skills supporting workforce development, developing policies and training programs**.

Introduction to job families

Job family refers to a **category of job roles** that perform related activities and have similar expertise across various organizations within a particular sector. Each **job family requires a distinct set of skills and tasks** to carry out critical functions within that sector.

Job families **are the foundation** to help categorize jobs into broader skill-based groups, making it easier to develop career paths and identify sector skills needs.

The following section of the **report presents a comprehensive value-chain analysis** for the overall sector, leveraging two primary value chains from the perspectives of service providers and passengers: (1) Logistics Services and Transportation (2) Travel. This analysis meticulously maps **30 Logistics Services and Transportation** job families across sector value chains. Through this mapping, the section highlights how the **job families contribute to the overall structure and functionality of sector**.

Logistics Services & Transportation value chain	Travel value chain
Transportation operations 	Taxi operations 
Courier, express, and parcel operations 	Bus operations 
Vehicle fleet maintenance (EV and Non- EV) 	Vehicle fleet maintenance (EV and Non- EV) 
Rail operations 	Rail operations 
Railway maintenance 	Railway maintenance 
Aviation maintenance 	Aviation maintenance 
Airline flight operations 	Airline flight operations 
Airport and ground operations 	Airport and ground operations 
Shipping management and operations 	Shipping management and operations 
Maritime and vessel maintenance 	Maritime and vessel maintenance 
Maritime services 	Sales and customer service 
Port operations 	Network planning 
Freight forwarding operations 	Health, safety, security, environment and quality 
Purchasing and procurement operations 	
Special product handling 	
Analysis and reporting 	
Demand forecasting 	
Warehouse and fulfilment center operations 	
E-logistics 	
Additional product services 	
Logistics solutioning 	
Sales and customer service 	
Network planning 	
Logistics process improvement 	
Engineering and technical service 	
Advanced maintenance 	
Health, safety, security, environment and quality 	
Regulatory compliance 	

30 key job families identified within the Logistics Services and Transportation sector

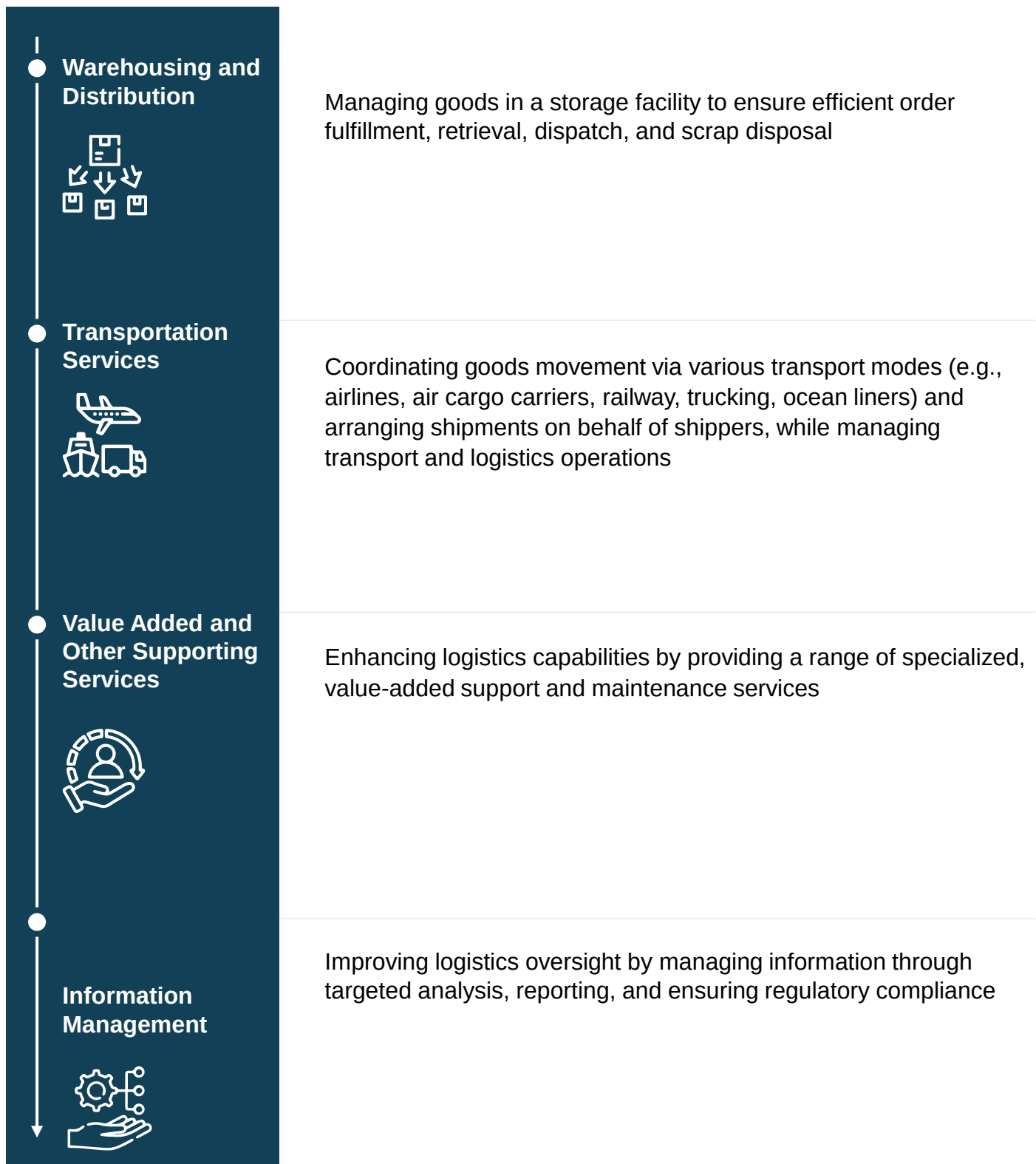
Taxi Operations 01 	Bus Operations 02 	Transportation Operations 03 
Courier, Express, and Parcel Operations 04 	Vehicle Fleet Maintenance (EV and non-EV) 05 	Rail Operations 06 
Railway Maintenance 07 	Aviation Maintenance 08 	Airline Flight Operations 09 
Airport and Ground Operations 10 	Shipping Management and Operations 11 	Maritime and Vessel Maintenance 12 
Maritime Services 13 	Port Operations 14 	Freight Forwarding Operations 15 

30 key job families identified within the Logistics Services and Transportation sector

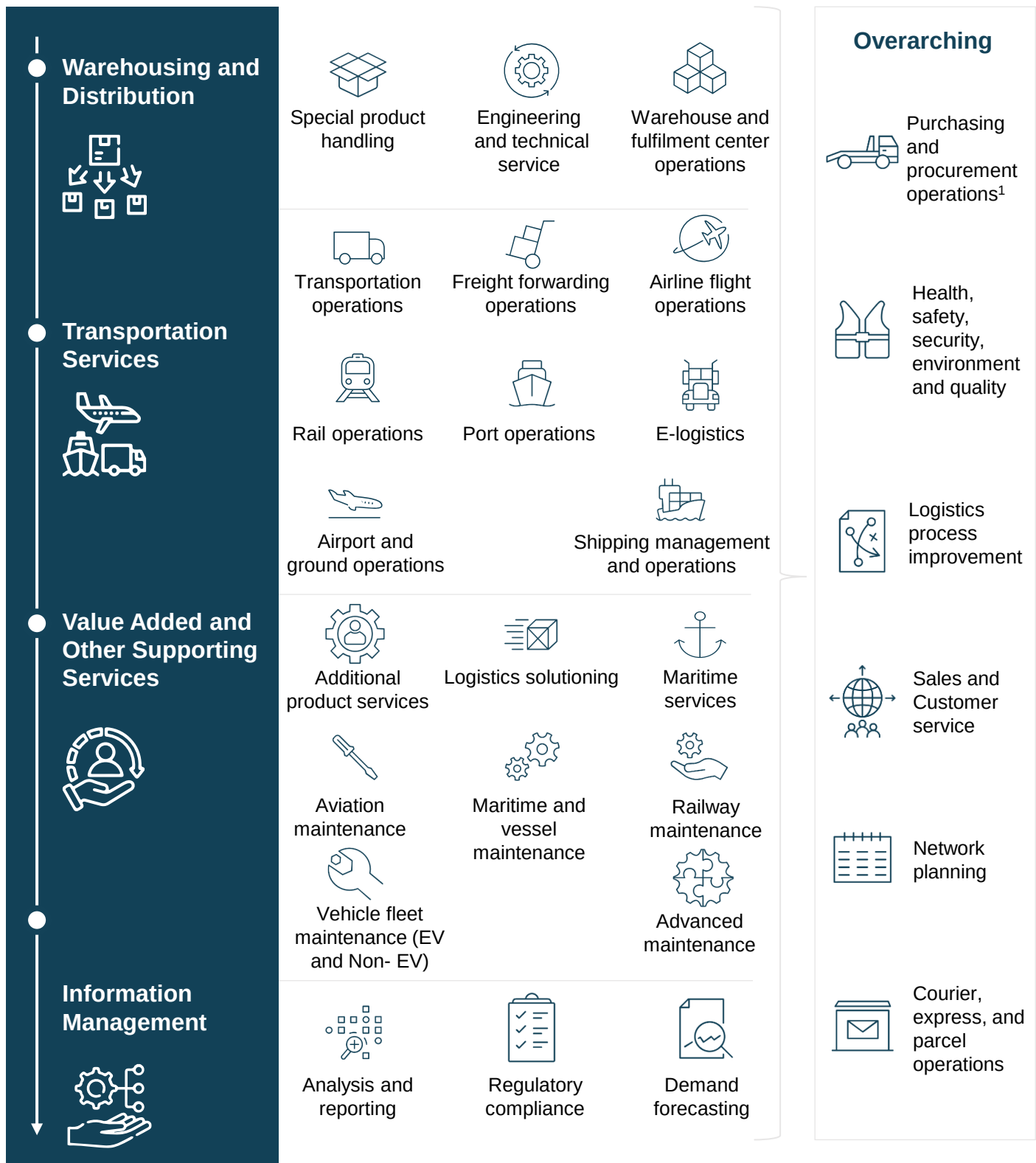
Purchasing and Procurement Operations 16 	Special Product Handling 17 	Analysis and Reporting 18 
Demand Forecasting 19 	Warehouse and Fulfilment Center Operations 20 	E-logistics 21 
Additional Product Services 22 	Logistics Solutioning 23 	Sales and Customer Service 24 
Network Planning 25 	Logistics Process Improvement 26 	Engineering and Technical Service 27 
Advanced Maintenance 28 	Health, Safety, Security, Environment and Quality 29 	Regulatory Compliance 30 

Logistics Services and Transportation value chain

The Logistics Services and Transportation value chain has 4 key elements: (1) Warehousing and distribution, (2) Transportation services, (3) Value added and other supporting services, (4) Information management



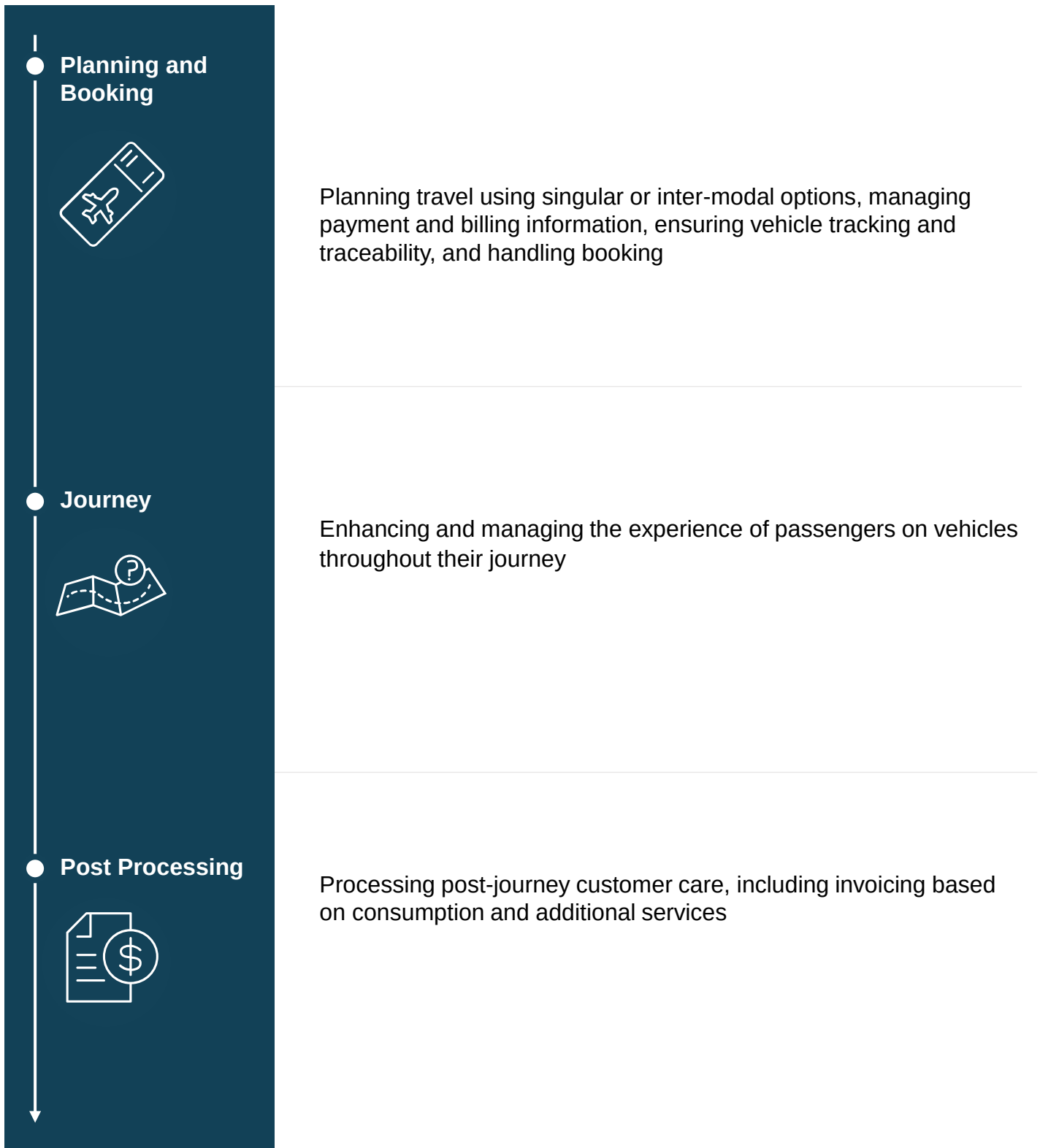
Job families mapped to the Logistics Services and Transportation value chain



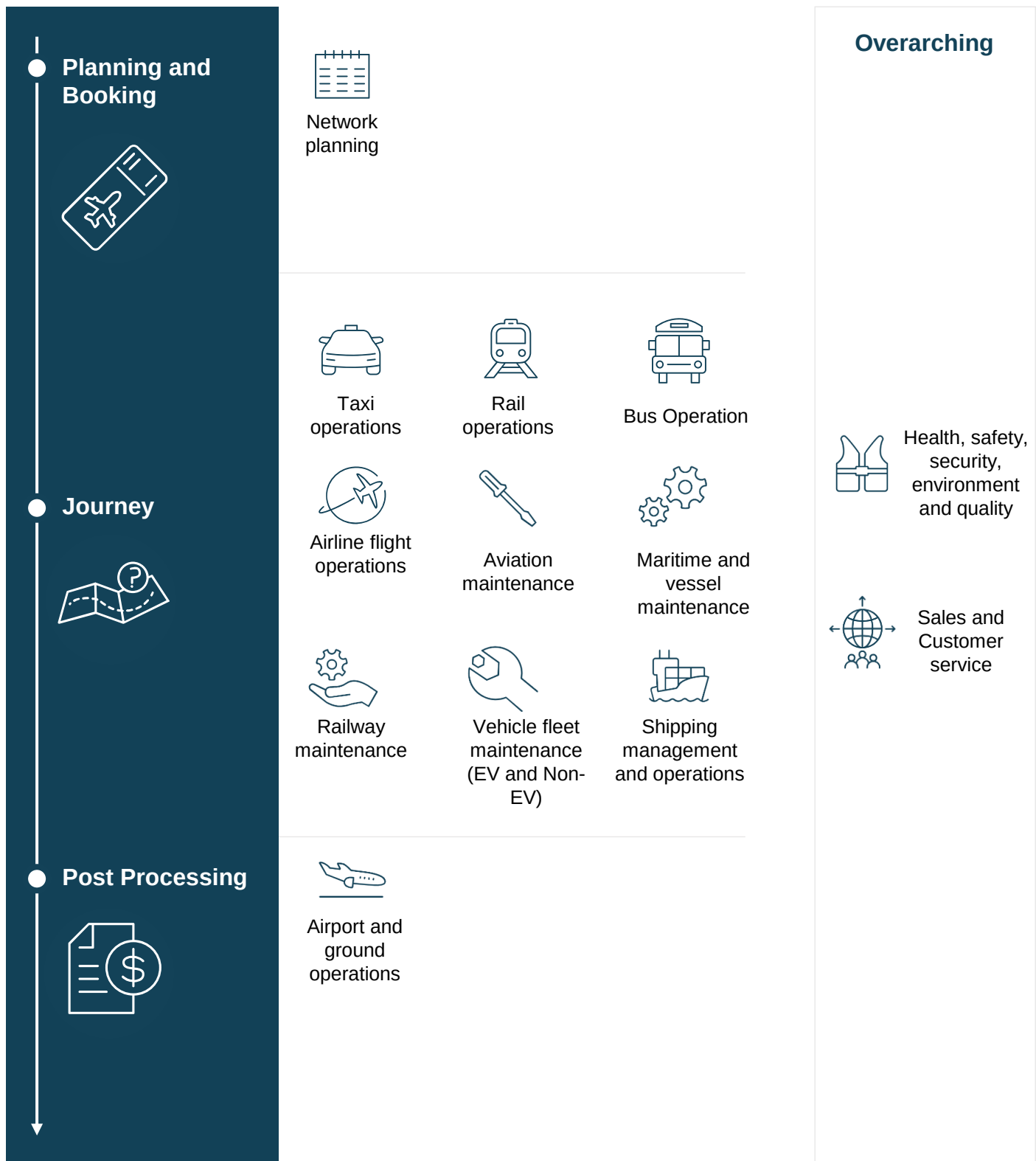
Note: Applicable to Warehousing and Distribution, Transportation Services, Value Added and Other Supporting Services

Travel value chain

The Travel value chain has 3 key elements: (1) Planning and Booking, (2) Journey, (3) Post processing



Job families mapped to the Travel value chain



Introduction to job family profiles

As part of the Sector Skills Framework, **job families have profiles accompanied with career pathway maps**. The profiles are essentially a guide to align job families and their skills requirements across a specific sector to ensure better workforce needs planning and development.

In the following section, job family profiles are detailed out for **30 job families** in the Logistics Services and Transportation sector. The **profiles outline in a structured way** the following elements:

- (1) Job family overview
- (2) Key job roles
- (3) Key tasks/ activities
- (4) Key relevant skills
- (5) Relevant certifications/ licenses
- (6) Relevant Unified Saudi Occupation Classification (USOC) codes

Job family overview	Offers a comprehensive brief of job family's role within the sector, emphasizing its core functions. It enables users to understand how the job family contributes to the broader sector
Key job roles	Provides a list of key roles in the job family enabling the users to understand what type of designations are available within the job family and what are the specific skills that are required for those jobs
Key tasks/ activities	Provides an overview of specific responsibilities and tasks associated with roles in the job family; helping users understand expectations
Key relevant skills	Provides list of essential skills for the roles in the job family to perform their key tasks as well as keep up with the requirements based on sector transformations
Relevant certifications/ licenses	Outlines select key licenses or certifications relevant (non-exhaustive) for key roles in the job family – in line with regulations/ policies in the Kingdom
Relevant USOC codes	Links relevant Unified Saudi Occupation Classification Codes on Unit level (level 4) to facilitate statistical reporting and automated updates

Logistics Services and Transportation Skills Dictionary Report and Logistics Services and Transportation Skills Repository are developed to complement this report. The dictionary provides detailed definitions on the sector's priority skills across the different proficiency levels, and the Excel tool is a repository of technical and general skills mapped to sub-sectors and job families.

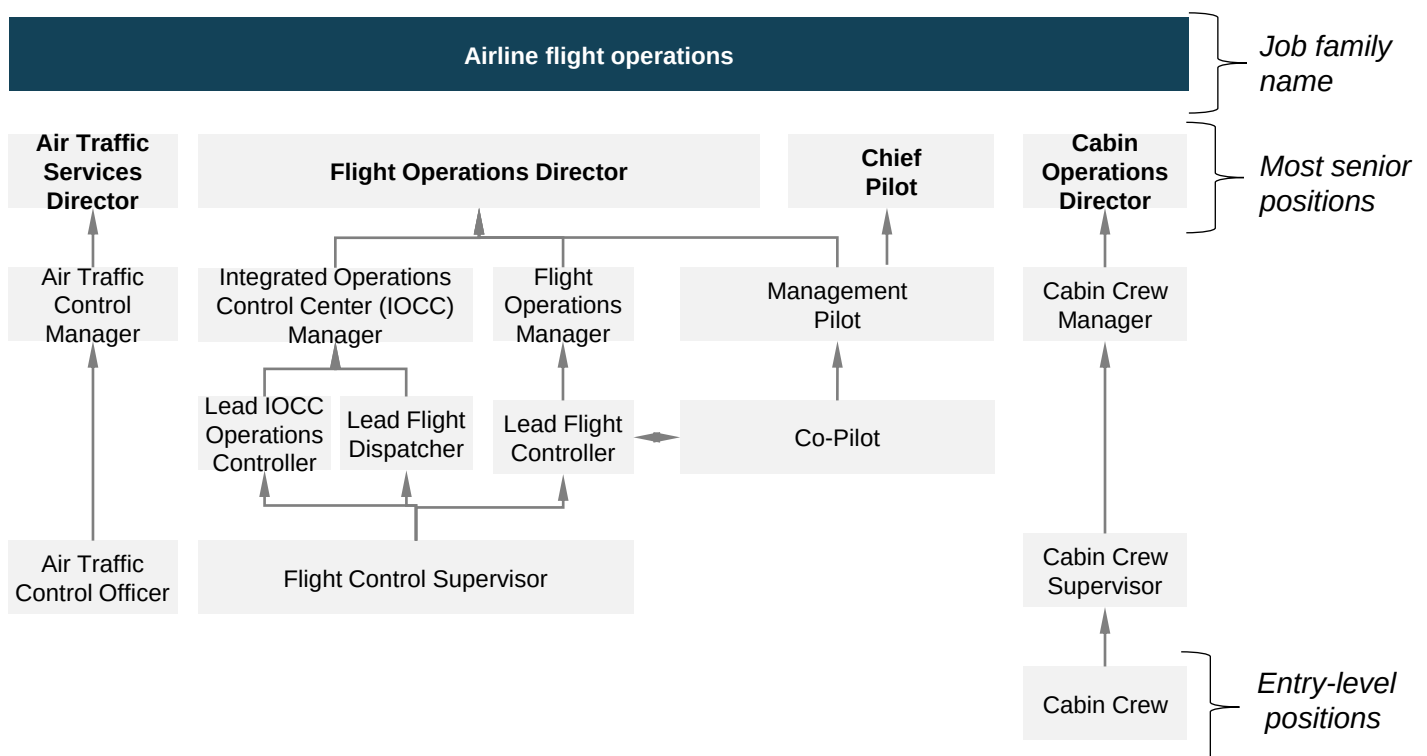
Introduction to career pathway maps

Based on the key roles identified across the 30 job families in the Logistics Services and Transportation sector, **20 career pathway maps have been developed covering 280+ job roles**. The pathways are **designed to identify current and future roles within the sector**, providing a **clear vision of how professionals can move on, progress and plan their careers within a job family**.

The **focus of these pathways is primarily on specialized technical areas** within the sector. This approach was taken to **target the roles needed** for the sector. While these pathways predominantly address technical functions, it is acknowledged that they do not cover other supporting roles.

The career pathway maps are a guide for sector stakeholders. Together with job family profiles, these career pathway maps provide an evidence-based foundation for designing relevant training programs, hiring strategies, educational curricula, skills initiatives, etc. These tools will help inform decisions on workforce development and ensure alignment with the industry's skill demands.

Illustrative example: Airline flight operations job family



280+ job roles identified across job families depicted in the profiles

Taxi Operations 01 	4 job roles	Bus Operations 02 	9 job roles	Transportation Operations 03 	11 job roles
Courier, Express, and Parcel Operations 04 	8 job roles	Vehicle Fleet Maintenance (EV and non-EV) 05 	8 job roles	Rail Operations 06 	13 job roles
Railway Maintenance 07 	23 job roles	Aviation Maintenance 08 	28 job roles	Airline Flight Operations 09 	17 job roles
Airport and Ground Operations 10 	24 job roles	Shipping Management and Operations 11 	14 job roles	Maritime and Vessel Maintenance 12 	4 job roles
Maritime Services 13 	26 job roles	Port Operations 14 	11 job roles	Freight Forwarding Operations 15 	8 job roles

280+ job roles identified across job families depicted in the profiles

Purchasing and Procurement Operations 16 	4 job roles	Special Product Handling 17 	5 job roles	Analysis and Reporting 18 	5 job roles
Demand Forecasting 19 	5 job roles	Warehouse and Fulfilment Center Operations 20 	8 job roles	E-logistics 21 	4 job roles
Additional Services 22 	4 job roles	Logistics Solutioning 23 	9 job roles	Sales and Customer Service 24 	8 job roles
Network Planning 25 	4 job roles	Logistics Process Improvement 26 	7 job roles	Engineering and Technical Service 27 	4 job roles
Advanced Maintenance 28 	6 job roles	Health, Safety, Security, Environment and Quality 29 	4 job roles	Regulatory Compliance 30 	4 job roles

Job family profiles and career pathway:

(1) Taxi Operations (2) Bus Operations

Job family overview

Taxi Operations and Bus Operations job families have closely related job roles **contributing to the overall management and operation of vehicle transportation services**. Given the synergies, the **two families are presented on a single career pathway map**.

Taxi Operations

Taxi Operations clusters role that deal with **managing and supervising the overall efficiency of taxi services**, ensuring timely transportation, safety, and customer satisfaction by overseeing drivers and maintaining seamless coordination.

Key tasks/ activities include:

- Driving taxi
- Streamlining fleet scheduling
- Ensuring driver accreditation
- Maintaining operational compliance
- Monitoring performance and safety
- Optimizing route planning
- Managing customer bookings and inquiries
- Overseeing vehicle maintenance and inspections
- Implementing cost control and budgeting
- Coordinating dispatch and availability of taxis

The family **has 4 key job roles**:

Corporate leadership position (e.g., Director of transport operations, Director of hub operations, etc.), Taxi operations manager, Taxi operations supervisor, Taxi driver

Bus Operations

Bus Operations clusters role that deal with **overall management and coordination of bus services**, focusing on ensuring smooth and efficient service delivery while maintaining operational standards and addressing logistical challenges.

Key tasks/ activities include:

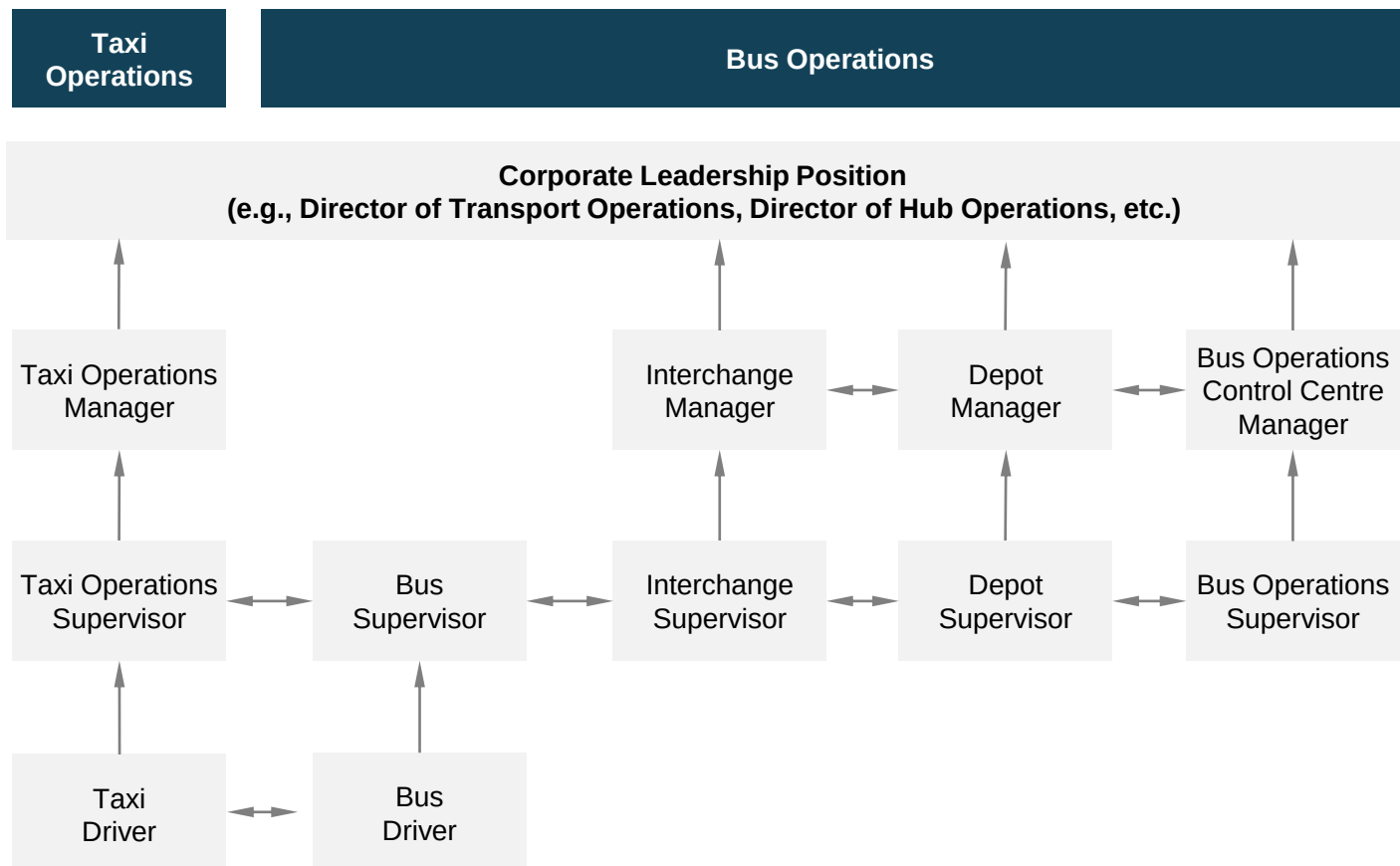
- Driving the bus
- Optimizing bus routes
- Scheduling driver availability
- Collecting fares and managing ticketing systems
- Responding to and managing incidents
- Monitoring real-time traffic

The family **has 9 key job roles**:

Corporate leadership position (e.g., Director of transport operations, Director of hub operations, etc.), Interchange manager, Depot manager, Bus operations control center manager, Bus supervisor, Interchange supervisor, Depot supervisor, Bus operations supervisor, Bus Driver

Job family profiles and career pathway: (1) Taxi Operations (2) Bus Operations

Vehicle fleet operations (*related job families presented together*)



Relevant USOC unit-tier codes for the job family: 1120 Managing directors and chief executives, 1324 Supply, distribution and related managers, 5112 Transport conductors, 8331 Bus and tram drivers, 8322 Car, taxi and van drivers

Skills requirements

The listed job roles require specialized skills related to taxi and bus services operations. The main skills for Taxi Operations include fleet management system, accident and incident response management, transport cost optimization, route planning etc. Similarly, the main skills for Bus Operations include accident and incident response management, bus service route operations training and development, bus vehicle operations, etc.

Full list of key skills is present in the following tables (see next page) providing a detailed view on job families' skills needs.

Job family profiles and career pathway: (1) Taxi Operations (2) Bus Operations

Key relevant skills: Taxi Operations

- Analysis of transport business networks
- Customer service delivery
- Fleet management system
- Light and medium vehicle operation
- Route planning and coordination
- Strategy development and planning
- Transport arrangement and planning
- Transport cost optimization
- Transportation route optimization
- Vehicle parking operations
- Workplace traffic safety management
- Safe and efficient driving

Key relevant skills: Bus Operations

- Accident and incident response management
- Bus service route operations training and development
- Bus vehicle operations
- Customer and passenger handling and care
- Interchange operations
- Vehicle parking operations
- Pricing for transportation services and operations
- Route planning and operations
- Transport cost optimization
- Safety and inspection
- Workplace health, safety and security procedures
- Transport planning and coordination
- Transportation route optimization

Relevant certifications/ licenses (non-exhaustive)

Both the job families require expertise in **vehicle operation and management**. In the Kingdom, for operation of taxi and bus services, **Taxi Routing and Bus Routing licenses issued by Transport General Authority** are mandatory. Additionally, taxi and bus drivers are required by law to hold valid Driver's license issued by the Ministry of Interior. Other than these licenses, there are also various acknowledged certifications and registrations for roles (mentioned below), that are prevalent in the Kingdom.

Certifications: Taxi Operations

- Taxi registration through Naql portal
- Road Safety Professional (RSP) Certification (Level 1) by Transportation Professional Certification Board

Certifications: Bus Operations

- Bus registration through Ministry of Interior
- Road Safety Professional (RSP) Certification (Level 1) by Transportation Professional Certification Board

Job family profiles and career pathway maps:

(3) Transportation Operations

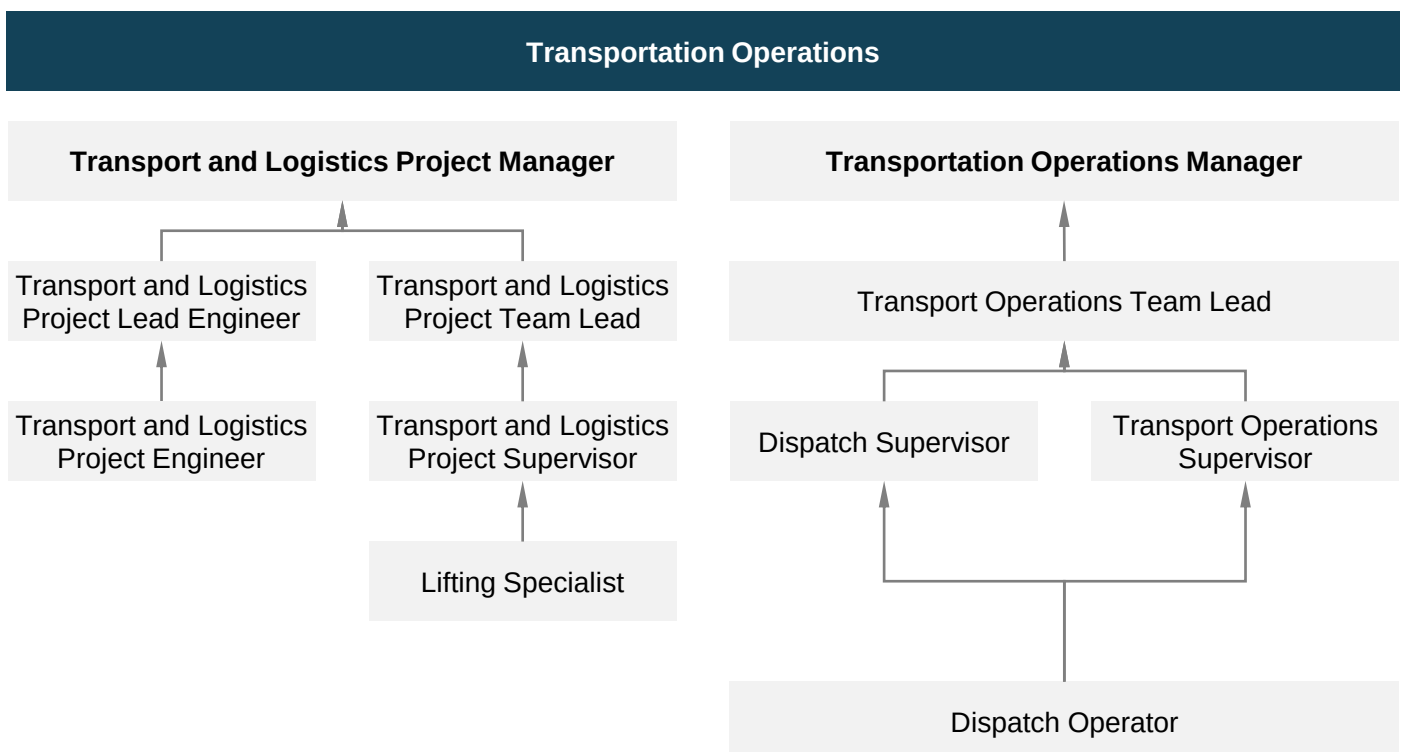
Job family overview

Transportation Operations job family clusters roles that focus on **managing day-to-day operations of road transportation and ensuring efficient movement of freight** through smooth coordination.

Key tasks/ activities include:

- Managing cross docking operations
- Coordinating freight transportation
- Supervising driving and loading/unloading processes
- Operating mobile cranes
- Overseeing warehouse coordination
- Tracking and monitoring shipments

The family has **11 key job roles**: Transport and logistics project manager, Transportation operations manager, Transport and logistics project lead engineer, Transport and logistics project team lead, Transport operations team lead, Transport and logistics project engineer, Transport and logistics project supervisor, Dispatch supervisor, Transport operations supervisor, Lifting specialist, Dispatch operator.



Relevant USOC unit-tier codes for the job family: 1324 Supply, distribution and related managers, 1213 Policy and planning managers, 8344 Lifting truck operators

Job family profiles and career pathway maps: (3) Transportation Operations

Skills requirements

The listed job roles require specialized skills in land transport and logistics operations. The main skills include bulk cargo operation, cargo handling and stowage, delivery management, goods packaging and labelling, last-mile delivery optimization, etc.

Full list of key skills is present in the following table providing a detailed view on job family skills needs.

Key relevant skills

- Bulk cargo operations
- Bulk transfer equipment usage
- Cargo handling and stowage
- Cargo transit and transshipment management
- Delivery management
- Goods distribution practices
- Goods packaging and labelling
- Inventory management
- Last-mile delivery optimization
- Post logistic activities
- Route planning and operations

Relevant certifications/ licenses (non-exhaustive)

The job family roles require strong knowledge of **transport operations management**. There are various certifications for roles (mentioned below) that are prevalent in the Kingdom.

Certifications

- International Certificate in Logistics and Transport (CILT)
- Certified in Logistics, Transportation, and Distribution (CLTD) through American Production and Inventory Control Society (APICS)

Job family profiles and career pathway maps: (4) Courier, Express, and Parcel Operations

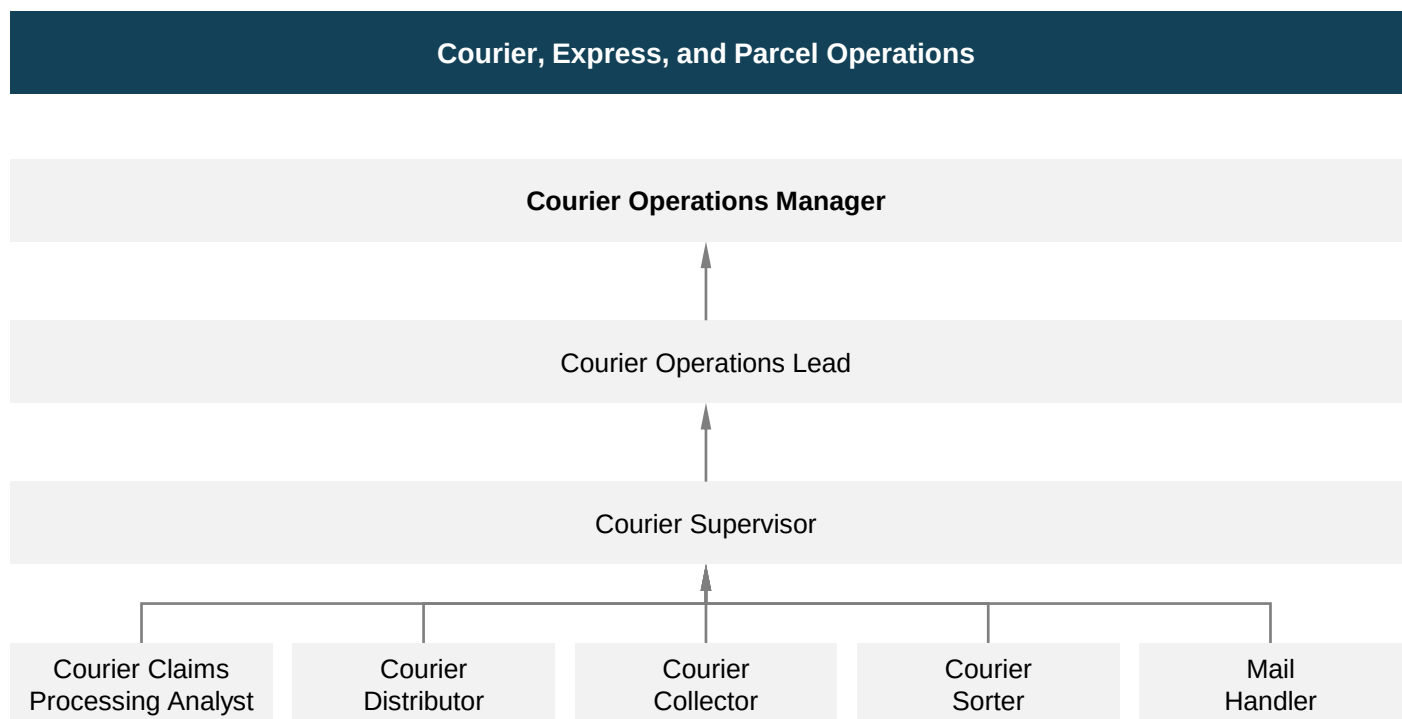
Job family overview

Courier, Express, and Parcel Operations clusters roles that focus on **executing timely and accurate delivery of packages with dedicated door-to-door services** with reliable and seamless handling of packages throughout the delivery process.

Key tasks/ activities include:

- Coordinating delivery schedules
- Managing sorting operations
- Planning and optimizing routes
- Package sorting and categorization
- Tracking and monitoring

The family has **8 key job roles**: Courier operations manager, Courier operations lead, Courier claims processing analyst, Courier supervisor, Courier distributor, Courier collector, Courier sorter, Mail handler.



Relevant USOC unit-tier codes for the job family: 4323 Transport clerks, 4412 Mail carriers and sorting clerks, 9621 Messengers, package deliverers and luggage porters

Job family profiles and career pathway maps: (4) Courier, Express, and Parcel Operations

Skills requirements

The listed job roles require skills in parcel handling and delivery operations. The main skills include order fulfilment administration, hub operations, product and goods inspection, transport cost optimization, etc.

Full list of key skills is present in the following table providing a detailed view on job family skills needs.

Key relevant skills

- Order fulfilment administration
- Customs clearance and procedures
- Delivery management
- Goods packaging and labelling
- Hub operations
- Inventory management
- Mails and parcels handling
- Order fulfilment administration
- Product and goods inspection
- Transport cost optimization
- Transportation route optimization

Relevant certifications/ licenses (non-exhaustive)

The job family roles require expertise in **delivery operations and management**. Prevalent certification for the listed roles in the Kingdom is given below.

Certifications

- Certified in Logistics Transportation, and Distribution (CLTD) through American Production and Inventory Control Society (APICS)

Job family profiles and career pathway maps: (5) Vehicle Fleet Maintenance (EV and non-EV)

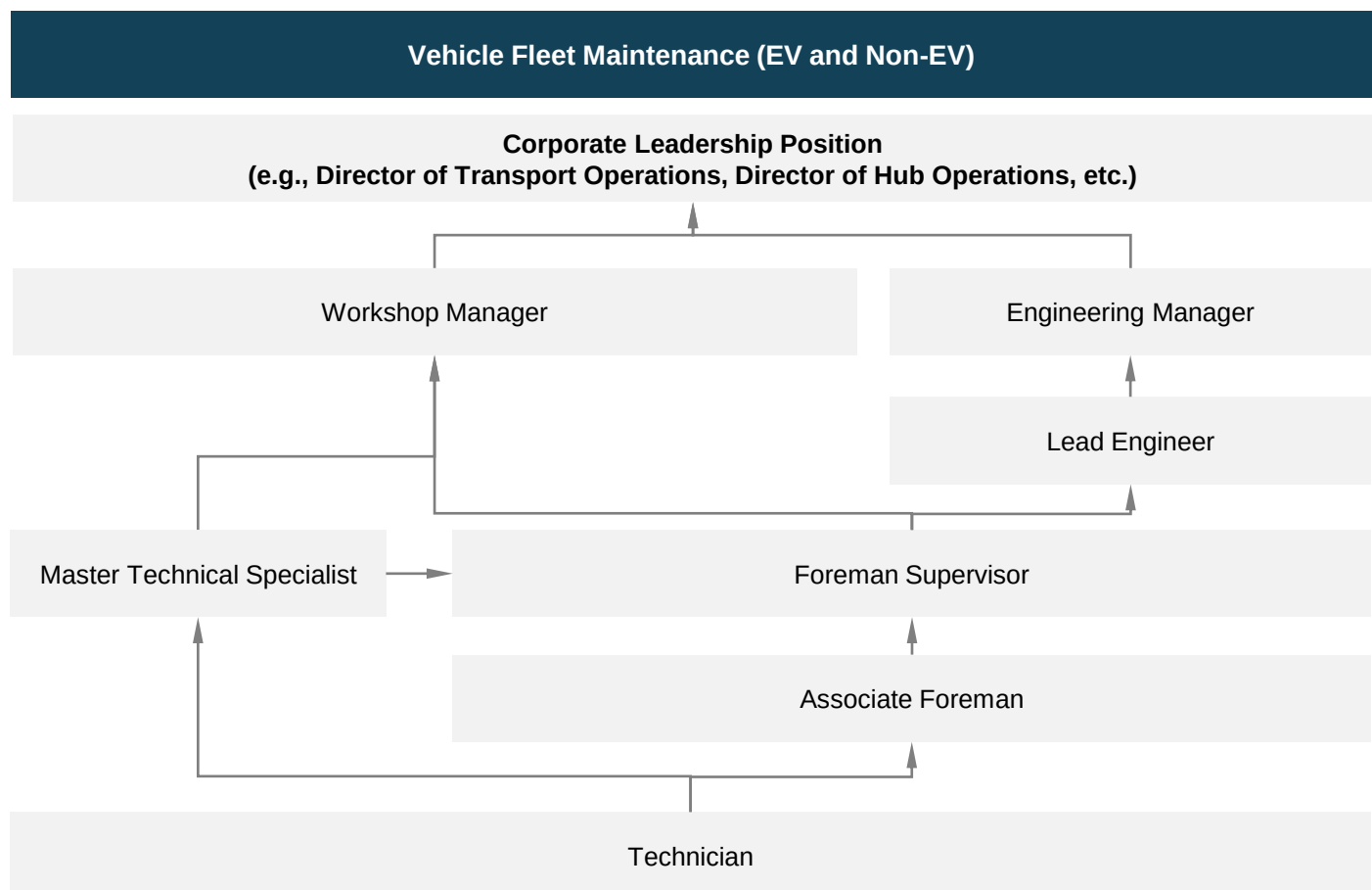
Job family overview

Vehicle Fleet Maintenance (EV and Non-EV) cluster roles that focus on **providing maintenance for a diverse fleet of vehicles, encompassing electric vehicles (EVs) and traditional combustion engines.**

Key tasks/activities include:

- Implementing and maintaining mobility platforms
- Managing EV charging station infrastructure
- Providing technical expertise in EV battery systems
- Providing maintenance services for traditional and non-traditional vehicles

The family has **8 key job roles**: Corporate leadership position (e.g., Director of transport operations, Director of hub operations, etc.), Workshop manager, Engineering manager, Lead Engineer, Master Technical Specialist, Foreman supervisor, Associate Foreman, Technician



Relevant USOC unit-tier codes for the job family: 1120 Managing directors and chief executives, 2151 Electrical engineers, 3113 Electrical engineering technicians, 3115 Mechanical engineering technicians, 7231 Motor vehicle mechanics and repairers, 7412 Electrical mechanics and fitters, 2144 Mechanical engineers, 8211 Mechanical machinery assemblers

Job family profiles and career pathway maps: (5) Vehicle Fleet Maintenance (EV and non-EV)

Skills requirements

The listed job roles require specialized skills in vehicle repair and maintenance. The main skills include computer-aided trouble-shooting, image processing and industrial vision inspection, process improvement and optimization, vehicle maintenance and cleanliness, etc.

Full list of key skills is present in the following table providing a detailed view on job family skills needs.

Key relevant skills

- Computer-aided trouble-shooting (automotive)
- Geometric dimensioning and tolerancing
- Image processing and industrial vision inspection
- Mechanical Aptitude
- Operation of Lubrication Equipment
- Petrol engine reconditioning
- Predictive maintenance
- Process improvement and optimization
- Precision measurement
- Vehicle maintenance and cleanliness

Relevant certifications/ licenses (non-exhaustive)

The job family roles require strong engineering expertise with a **significant presence of mechanical and electrical engineers and technicians** for the roles. In the Kingdom, practicing engineers and technicians are required by law to hold **membership with the Saudi Council of Engineers (SCE)**, ensuring that professionals meet national standards and adhere to engineering regulations. In addition to this, there are also highly acknowledged certifications for roles (mentioned below), that is prevalent in the Kingdom.

Certifications

- Fundamentals of Engineering Certificate (FE) by National Council of Examiners for Engineering and Surveying

Job family profiles and career pathway maps: (6) Rail Operations

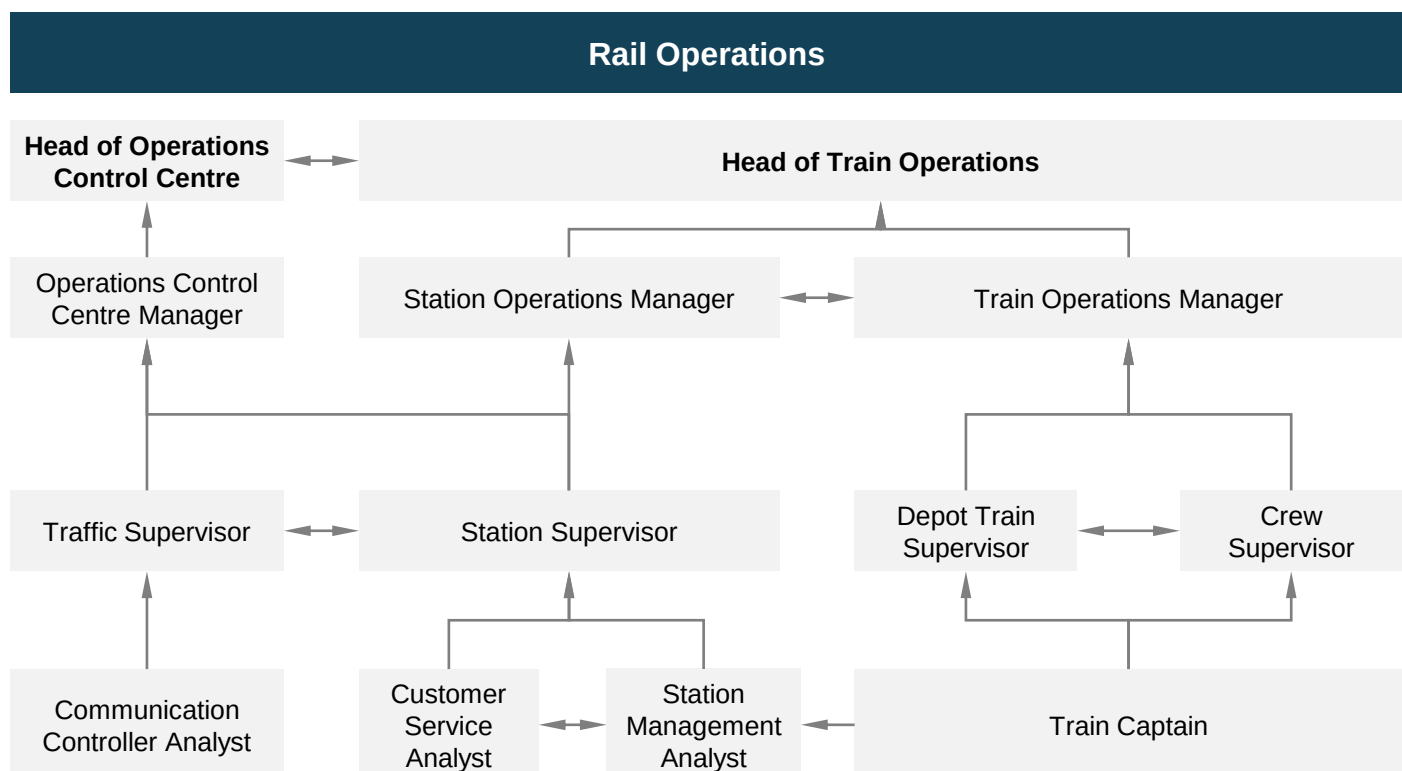
Job family overview

Rail Operations job family clusters roles that focus on **overseeing and coordinating of train movements** across a rail network, ensuring safety and efficiency for smooth transportation of both passengers and freight across various routes

Key tasks/activities include:

- Train and network controlling
- Managing rail signaling systems
- Operating light rail systems
- Driving, Handling disruptions
- Planning capacity
- Monitoring real-time traffic

The family has **13 key job roles**: Head of operations control centre, Operations control centre manager, Traffic supervisor, Communication controller analyst, Head of train operations, Station operations manager, Station supervisor, Station management analyst, Train operations manager, Depot train supervisor, Crew supervisor, Train captain.



Relevant USOC unit-tier codes for the job family: 1324 Supply, distribution and related managers, 3139 Process control technicians not elsewhere classified, 8311 Locomotive engine drivers, 8312 Railway brake, signal and switch operators

Job family profiles and career pathway maps: (6) Rail Operations

Skills requirements

The listed job roles require specialized skills in rail locomotive and general operations. The main skills include accident and incident response management, awareness of railway fundamentals, engineering train and rolling stock operation, locomotive operations, rail freight organization and safety, rail network technical support, etc.

Full list of key skills is present in the following table providing a detailed view on job family skills needs.

Key relevant skills

- Accident and incident response management
- Awareness of railway fundamentals
- Engineering train and rolling stock operation
- Locomotive operations
- Rail freight organization and safety
- Rail network technical support
- Rail yard management
- Route planning and coordination
- Signaling structures and equipment
- Train depot control management
- Train fare and ticket management

Relevant certifications/ licenses (non-exhaustive)

The job family's roles require expertise in **rail operations and management**. In the Kingdom, **license to Transport Passengers and Freight** is mandatory for rail operation services. Additionally, locomotive drivers are required by law to **hold Driver's license issued through Saudi Railway Polytechnic**. Other than these licenses, there are also various acknowledged certifications for roles (mentioned below), that are prevalent in the Kingdom.

Certifications

- Rail driver trainings through Saudi Railway Polytechnic (SRP)
- Rail driver diploma program through Saudi Railway Polytechnic (SRP)

Job family profiles and career pathway maps:

(7) Railway Maintenance

Job family overview

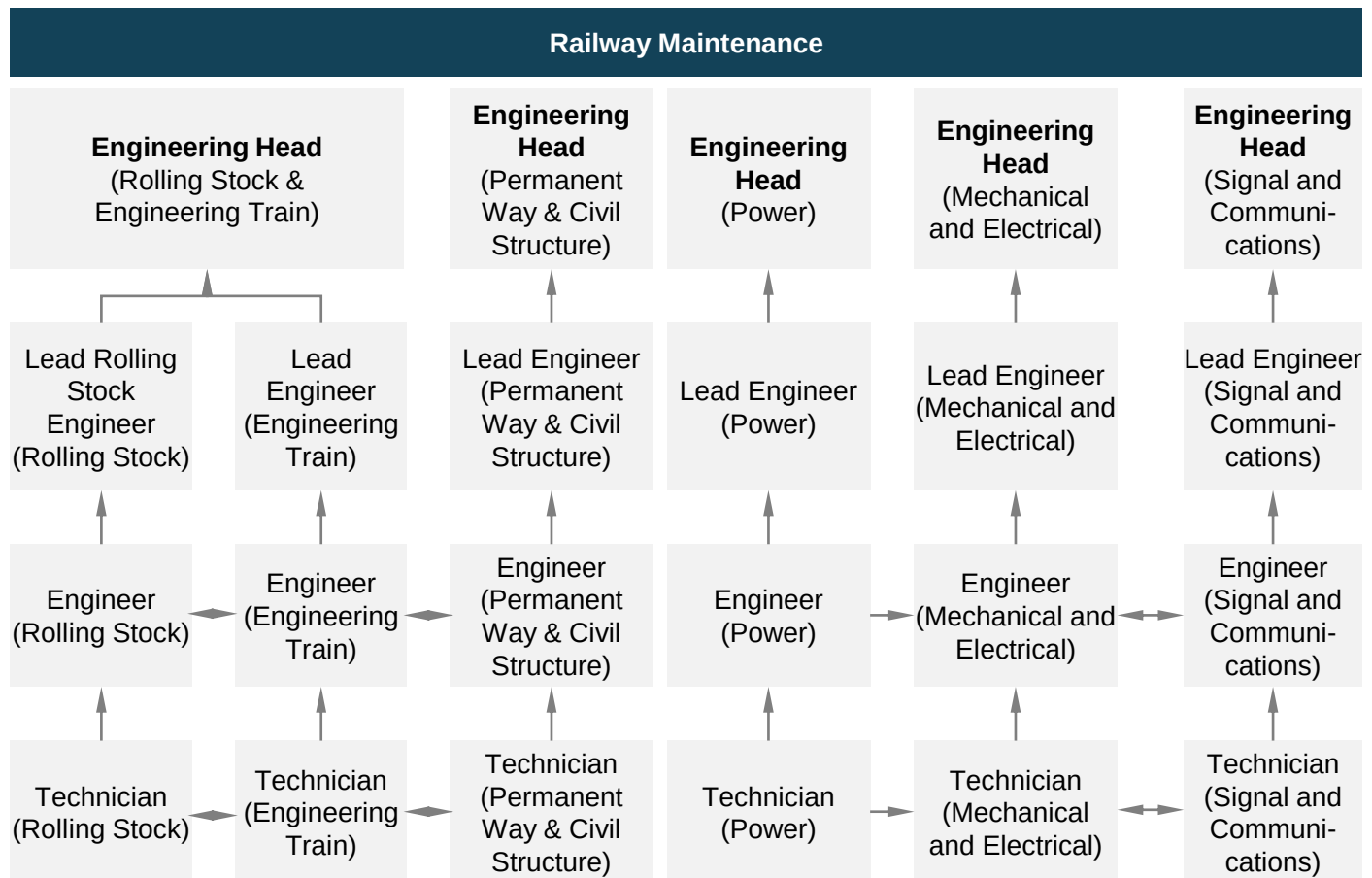
Railway Maintenance job family clusters roles that deal with regular **maintenance of rolling stock and railway systems, ensuring the functionality of infrastructure like tracks, signaling, and communication networks** for safe and uninterrupted operations.

Key tasks/ activities include:

- Routine maintenance of tracks, signals, and telecommunications
- Servicing rolling stock
- Ensuring system integrity
- Maintaining light rail systems
- Risk assessment and mitigation

The family has **23 key job roles**: Engineering head (Rolling stock & Engineering train), Lead engineer (rolling stock), Engineer (rolling stock), Technician (rolling stock), Lead engineer (Engineering Train), Engineer (Engineering Train), Technician (Engineering Train), Engineering head (Permanent Way & Civil Structure), Lead engineer (Permanent Way & Civil Structure), Engineer (Permanent Way & Civil Structure), Technician (Permanent Way & Civil Structure), Engineering head (Power), Lead engineer (Power), Engineer (Power), Technician (Power), Engineering head (Mechanical & Electrical), Lead engineer (Mechanical & Electrical), Engineer (Mechanical & Electrical), Technician (Mechanical & Electrical), Engineering head (Signal & Communication), Lead engineer (Signal & Communication), Engineer (Signal & Communication), Technician (Signal & Communication).

Job family profiles and career pathway maps: (7) Railway Maintenance



Relevant USOC unit-tier codes for the job family: 2144 Mechanical engineers, 2151 Electrical engineers, 2153 Telecommunication Engineers, 3115 Mechanical engineering technicians, 7412 Electrical mechanics and fitters, 3113 Electrical engineering technicians

Skills requirements

The listed job roles require specialized skills related to railway infra and locomotive maintenance. The main skills include autonomous systems engineering and maintenance, communication power system maintenance, engineering train and rolling stock operation, rail equipment servicing, rail installation and maintenance, rail track maintenance, etc.

Full list of key skills is present in the following table (see next page) providing a detailed view on job family skills needs.

Job family profiles and career pathway maps: (7) Railway Maintenance

Key relevant skills

- Autonomous systems engineering and maintenance
- Communication power system maintenance
- Engineering train and rolling stock operation
- Rail equipment servicing
- Rail installation and maintenance
- Rail track maintenance
- Rail yard management
- Railway safety and risk assessment
- Train brake systems maintenance
- Vehicle and equipment inspection skills

Relevant certifications/ licenses (non-exhaustive)

The job family roles require strong engineering expertise with a **significant presence of mechanical, railway and electrical engineers and technicians** for the roles. In the Kingdom, practicing engineers and technicians are required by law to hold **membership with the Saudi Council of Engineers (SCE)**, ensuring that professionals meet national standards and adhere to engineering regulations. In addition to this, there are also various highly acknowledged certifications for roles (mentioned below), that are prevalent in the Kingdom.

Certifications

- Fundamentals of Engineering Certificate (FE) by National Council of Examiners for Engineering and Surveying
- Rail maintenance trainings and diploma by Saudi Railway Polytechnic (SRP)
- Certified Maintenance and Reliability Professional (CMRP) credential by Society for Maintenance and Reliability Professionals

Job family profiles and career pathway maps:

(8) Aviation Maintenance

Job family overview

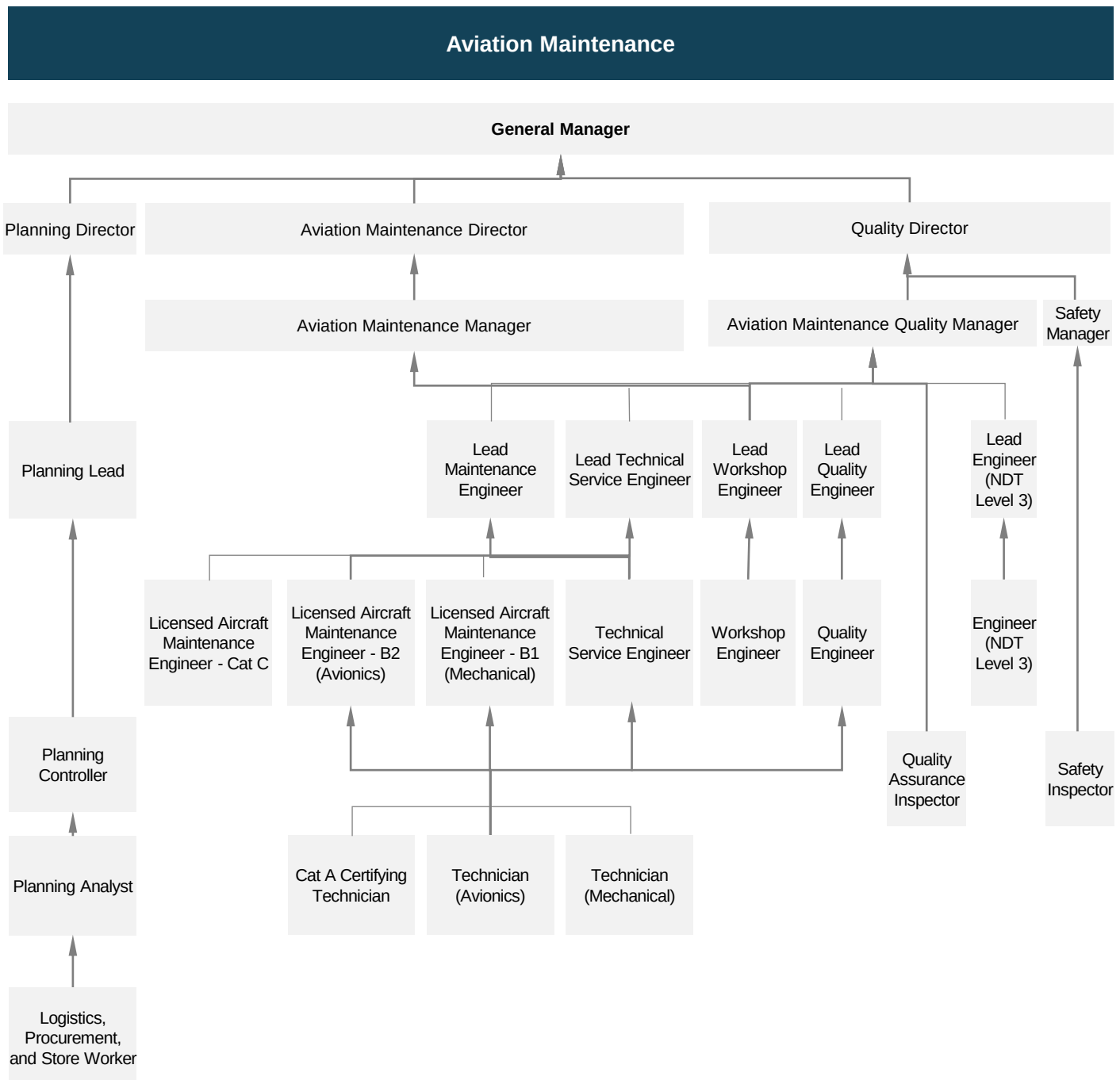
The job family clusters roles that focus on **systematic inspection, repair, and overhaul of aircrafts** through rigorous processes to guarantee airworthiness, while adhering to regulatory standards.

Key tasks/activities include:

- Executing preventive aircraft maintenance
- Servicing aircraft components
- Undertaking structural integrity checks
- Diagnosing electronic and mechanical systems
- Managing general repairs

The family has **28 key job roles**: General manager, Aviation maintenance director, Quality Director, Planning director, Aviation maintenance manager, Quality manager, Lead maintenance engineer, Lead technical service engineer, Lead workshop engineer, Lead quality engineer, Lead engineer (NDT Level 3), Planning lead, Licensed aircraft maintenance engineer - Cat C, Licensed aircraft maintenance engineer - B1 (Mechanical), Licensed aircraft maintenance engineer - B2 (Avionics), Technical service engineer, Workshop engineer, Quality engineer, Engineer (NDT Level 3), Planning controller, Quality assurance inspector, Safety manager, Safety inspector, Cat A certifying technician, Technician (Avionics), Technician (Mechanical), Planning analyst, Logistics, procurement, and store worker.

Job family profiles and career pathway maps: (8) Aviation Maintenance



Relevant USOC unit-tier codes for the job family: 1120 Managing directors and chief executives, 1324 Supply, distribution and related managers, 2144 Mechanical engineers, 3155 Air traffic safety electronics technicians, 3115 Mechanical engineering technicians, 7232 Aircraft engine mechanics and repairers, 7412 Electrical mechanics and fitters, 7421 Electronics mechanics and servicers, 1213 Policy and planning managers, 2149 Engineering professionals not elsewhere classified, 1213 Policy and planning managers, 2421 Management and organization analysts, 1349 Professional services managers not elsewhere classified,

Job family profiles and career pathway maps:

(8) Aviation Maintenance

Skills requirements

The listed job roles require specialized skills in aviation maintenance engineering. The main skills include aerospace maintenance practices application, aircraft maintenance (MRO), automation design, autonomous systems engineering and maintenance, etc.

Full **list of key skills is present in the following table** providing a detailed view on job family skills needs.

Key relevant skills

- | | |
|--|--|
| • Aerospace maintenance practices application | • Customer experience management |
| • Aircraft maintenance (MRO) | • Mechanical mounting |
| • Automation Design | • Process improvement and optimization |
| • Autonomous systems engineering and maintenance | • Maintenance scheduling |
| • Electrical systems (aircraft) | • Technology integration and application |

Relevant certifications/ licenses (non-exhaustive)

The job family roles require strong engineering expertise in aeronautical engineering and **mechanical, aerospace, aeronautical, and electrical engineers and technicians** for the roles. In the Kingdom, practicing engineers and technicians are required by law to hold **membership with the Saudi Council of Engineers (SCE)**, ensuring that professionals meet national standards and adhere to engineering regulations. In addition to this, there are also various highly acknowledged certifications for roles (mentioned below), that are prevalent in the Kingdom.

Certifications

- Fundamentals of Engineering Certificate (FE) by National Council of Examiners for Engineering and Surveying
- Aircraft maintenance training courses by the National Aviation Academy (TAYARAN)
- International Air Transport Association certificate/diploma courses on safety management
- Aircraft Maintenance Technician license through General Authority of Civil Aviation (GACA)

Job family profiles and career pathway maps: (9) Airline Flight Operations

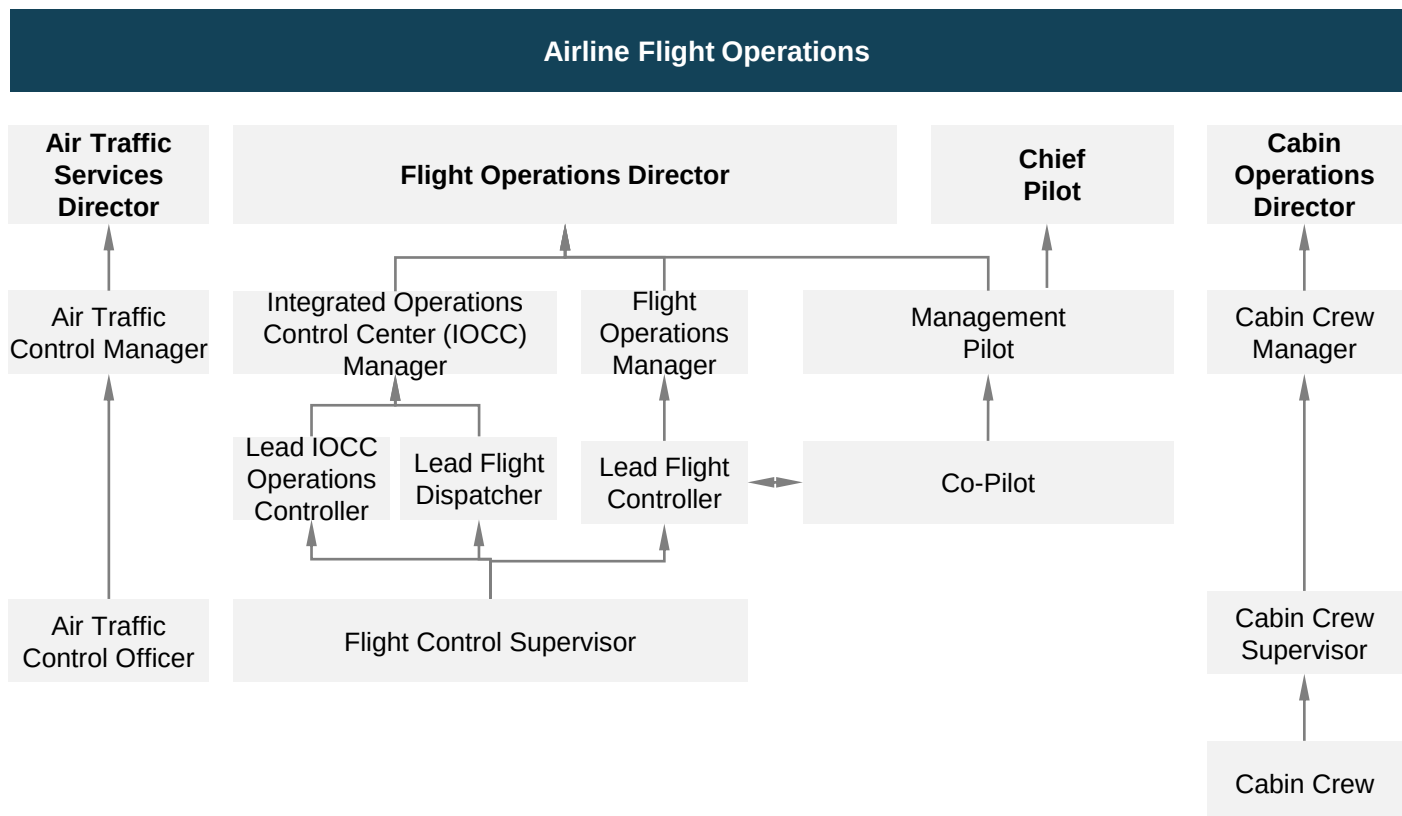
Job family overview

The job family clusters roles that focus on the **management and supervision of passenger and cargo flights**, ensuring safety, efficiency, regulatory compliance, and coordination from pre-flight preparations to post-flight procedures.

Key tasks/activities include:

- Piloting aircrafts
- Cabin attending
- Stewarding
- Coordinating with ground handling teams
- Communicating with and servicing passengers
- Managing and responding to incidents
- Ensuring compliance to flight safety regulations
- Tracking aircraft maintenance

The family has **17 key job roles**: Air traffic services director, Air traffic control manager, Air traffic control officer, Chief pilot, Flight operations director, Cabin operations director, Management pilot, Flight operations manager, Integrated operations control center (IOCC) manager, Cabin crew Manager, Lead IOCC operations controller, Lead flight dispatcher, Lead flight controller, Co-pilot, Flight control supervisor, Cabin crew supervisor, Cabin crew.



Relevant USOC unit-tier codes for the job family: 1324 Supply, distribution and related managers, 3153 Aircraft pilots and related associate professionals, 5111 Travel attendants and travel stewards, 3154 Air traffic controllers

Job family profiles and career pathway maps:

(9) Airline Flight Operations

Skills requirements

The listed job roles require specialized skills related to flight services and aircraft operations. The main skills include aircraft systems knowledge, aircraft manual handling, airline operations management, crisis management and communication, food and beverage safety, hygiene and security, etc.

Full list of key skills is present in the following table providing a detailed view on job family skills needs.

Key relevant skills

- | | |
|--|--|
| • Aircraft Systems Knowledge | • Aircraft landing operations |
| • Aircraft manual handling | • Navigation & route identification |
| • Airline operations management | • Pilot communication |
| • Crisis management and communication | • Post-landing operations |
| • Food and beverage safety, hygiene and security | • Pre-flight preparation |
| | • Baggage loading and unloading administration |

Relevant certifications/ licenses (non-exhaustive)

The job family roles require expertise in **flight operations and management**. In Kingdom, all pilots are required by law to obtain Pilot's license through General Authority of Civil Aviation. In addition to this, there are also various highly acknowledged certifications for roles (mentioned below), that are prevalent in the Kingdom.

Certifications

- International Air Transport Association certificate/diploma courses on incident prevention, airside operations, ground operations
- Ground flight and airline flight instructor through General Authority of Civil Aviation (GACA)
- International Air Transport Association certificate/diploma courses on Advanced Diploma in Airport Management and Operation

Job family profiles and career pathway maps:

(10) Airport and Ground Operations

Job family overview

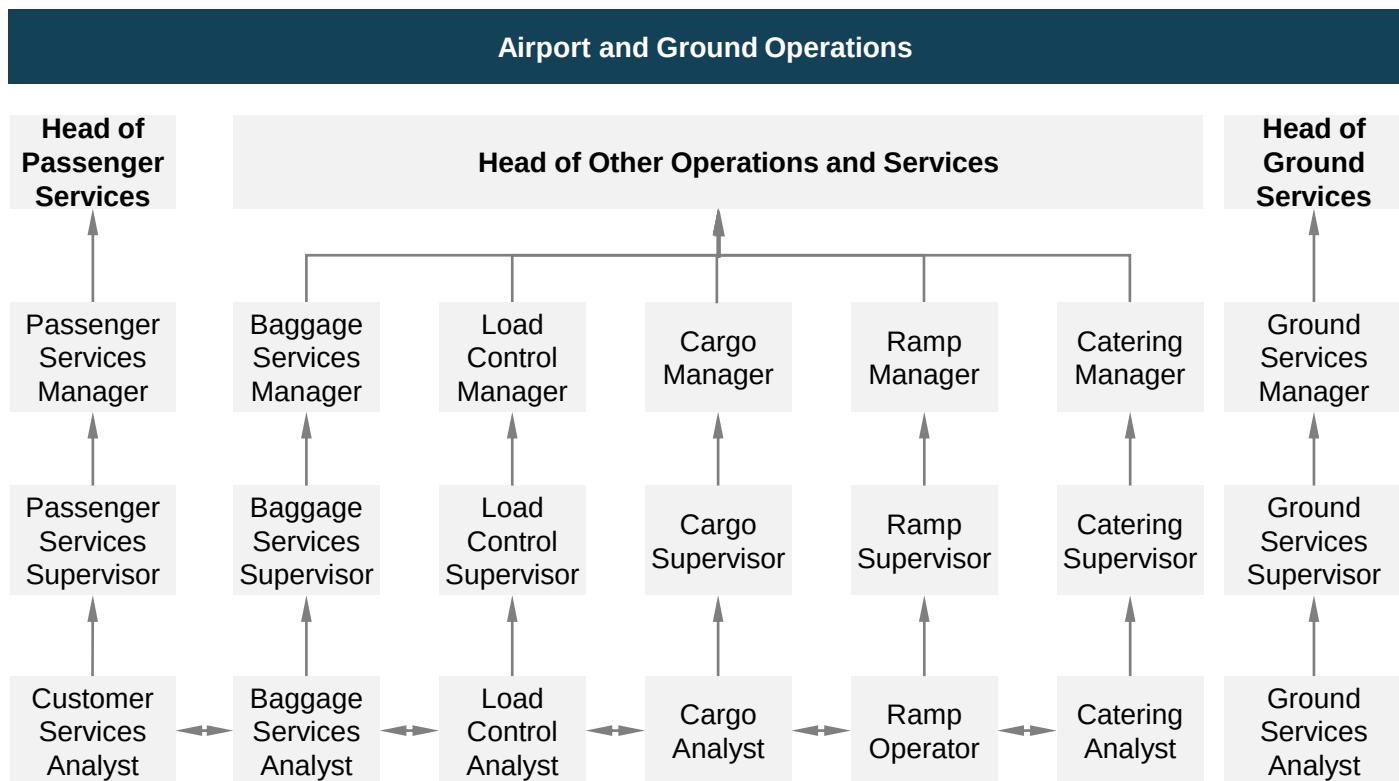
The job family clusters roles that focus on **streamlining operational efficiency in airport and airline services** through integrated management of all ground-related activities within the airport and airline domains.

Key tasks/activities include:

- Handling ramps
- Loading and Baggage servicing
- Catering
- Providing clerical support
- Managing air traffic
- Executing on-ground flight procedures
- Coordinating with in-flight operations
- Ensuring optimum load balance
- Providing on-ground technical services for aircrafts

The family has **24 key job roles**: Head of passenger services, Head of other operations and services, Head of ground services, Passenger services manager, Ground services manager, Baggage services manager, Load control manager, Cargo manager, Ramp manager, Catering manager, Passenger services supervisor, Ground services supervisor, Baggage services supervisor, Load control supervisor, Cargo supervisor, Ramp supervisor, Catering supervisor, Customer services analyst, Baggage services analyst, Load control analyst, Cargo analyst, Ramp operator, Catering analyst, Ground services analyst.

Job family profiles and career pathway maps: (10) Airport and Ground Operations



Relevant USOC unit-tier codes for the job family: 1324 Supply, distribution and related managers, 5111 Travel attendants and travel stewards, 9333 Freight handlers

Skills requirements

The listed job roles require specialized skills related to operations of ground services. The main skills include aircraft Equipment management (ULD), aircraft load planning, cargo handling and stowage, cargo transit and transshipment management, accident and incident response management, airport operations management etc.

Full list of key skills is present in the following table (see next page) providing a detailed view on job family skills needs.

Job family profiles and career pathway maps: (10) Airport and Ground Operations

Key relevant skills

- Aircraft Equipment Management (ULD)
- Aircraft load planning
- Cargo handling and stowage
- Cargo transit and transshipment management
- Accident and incident response management
- Airport operations management
- Airport service quality management
- Import and export documentation administration
- On-demand logistics coordination
- Signaling system

Relevant certifications/ licenses (non-exhaustive)

The job family roles require strong expertise in **airline ground operations and management**. There are various highly acknowledged certifications for roles (mentioned below), that are prevalent in the Kingdom.

Certifications

- Safety, compliance and airport operations trainings through Saudi Academy of Civil Aviation (SACA)
- Ground operations management trainings and diploma by International Air Transport Association
- Advanced Diploma in Airport Management and Operation by International Air Transport Association
- Aircraft Maintenance Technician license through General Authority of Civil Aviation (GACA)
- Air traffic controller License through General Authority of Civil Aviation (GACA)
- Ground Movement Coordinator License through General Authority of Civil Aviation (GACA)

Job family profiles and career pathway maps:

(11) Shipping Management and Operations

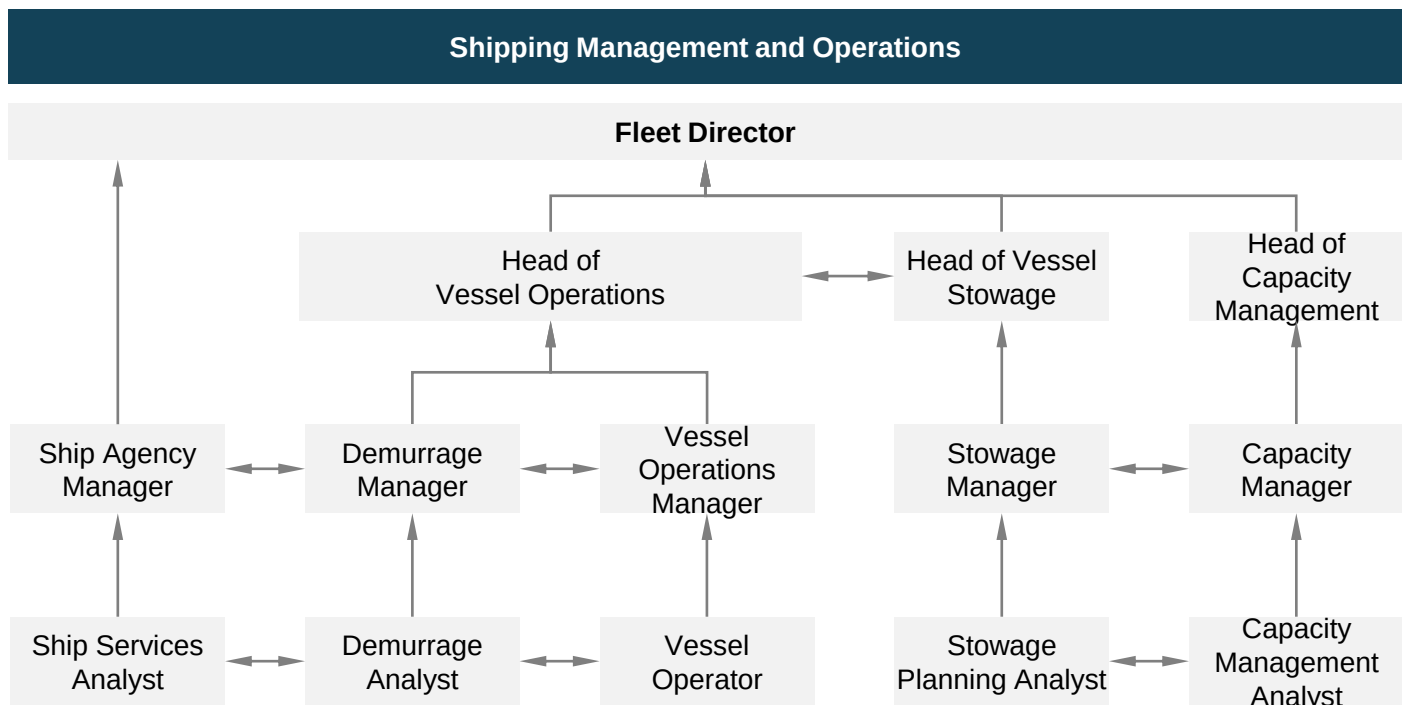
Job family overview

The job family clusters roles that focus on overseeing the **day-to-day functioning of both passenger and cargo vessels** including overseeing vessel movements, ensuring regulatory compliance, and coordinating with port authorities.

Key tasks/ activities include:

- Servicing passengers
- Undertaking Deck and Engine operations
- Managing fleet performance
- Carrying out drydocking operations
- Managing crew and cargo
- Voyage management
- Monitoring vessel movements
- Optimizing routes
- Ensuring compliance

The family has **14 key job roles**: Fleet director, Head of vessel operations, Head of vessel stowage, Head of capacity management, Vessel operations manager, Stowage manager, Capacity manager, Ship agency manager, Demurrage manager, Stowage analyst, Ship services analyst, Demurrage analyst, Vessel operator, Capacity management analyst.



Relevant USOC unit-tier codes for the job family: 1120 Managing directors and chief executives, 1324 Supply, distribution and related managers, 3152 Ships' deck officers and pilots

Job family profiles and career pathway maps:

(11) Shipping Management and Operations

Skills requirements

The listed job roles require specialized skills related to ship vessel operations and management. The main skills include cargo handling and stowage, container loading and unloading administration, crew management, maritime emergency response management, maritime incident management, navigation and route identification, etc.

Full list of key skills is present in the following table providing a detailed view on job family skills needs.

Key relevant skills

- Cargo handling and stowage
- Container loading and unloading administration
- Crew management
- Maritime emergency response management
- Maritime incident management
- Navigation & route identification
- Time sensitive cargo delivery management
- Shipping network operations management
- Shipping trade management
- Route planning and operations

Relevant certifications/ licenses (non-exhaustive)

The job family roles require expertise in **shipping operations and management**. In the Kingdom, **Maritime license through Saudi Ports Authority and Maritime license for Transportation of Individuals and Goods issued by Transport General Authority**, are required for marine fleet management and operations. In addition to this, shipping agents are required by law to hold a **Shipping Agent license issued by Zakat, Tax, and Customs Authority** for practicing the profession. Other than these licenses, there are also various highly acknowledged certifications and registrations for roles (mentioned below), that are prevalent in the Kingdom.

Certifications

- Ships and Maritime Units Registration with Transport General Authority, Saudi Arabia
- International Certificate in Logistics and Transport (CILT)

Job family profiles and career pathway maps:

(12) Maritime and Vessel Maintenance (13) Maritime Services

Job family overview

Maritime and Vessel Maintenance and Maritime Services job families have **closely related job roles** contributing to the overall **maritime management, maintenance and services**. Given the synergies, the **two families are presented on a single career pathway map**.

Maritime and Vessel maintenance

The job family clusters roles that focus on **ensuring the operational readiness, safety, and longevity of various types of maritime vessels**, including cargo ships, tankers, and passenger liners.

Key tasks/activities include:

- Overseeing hull
- Machinery and equipment servicing
- Ensuring the reliability of navigation and communication systems
- Maintaining electrical and electronic systems
- Conducting safety equipment checks

The family **has 4 key job roles**:

General manager, Maritime and vessel maintenance technical manager, Maritime and vessel maintenance technical officer, Maritime and vessel maintenance technician

Maritime Services

The job family clusters roles that focus on **offering a wide range of support services that are crucial for the efficient and safe operation of the shipping industry**, both commercial and governmental maritime operations, including cargo transportation, offshore services, and passenger travel.

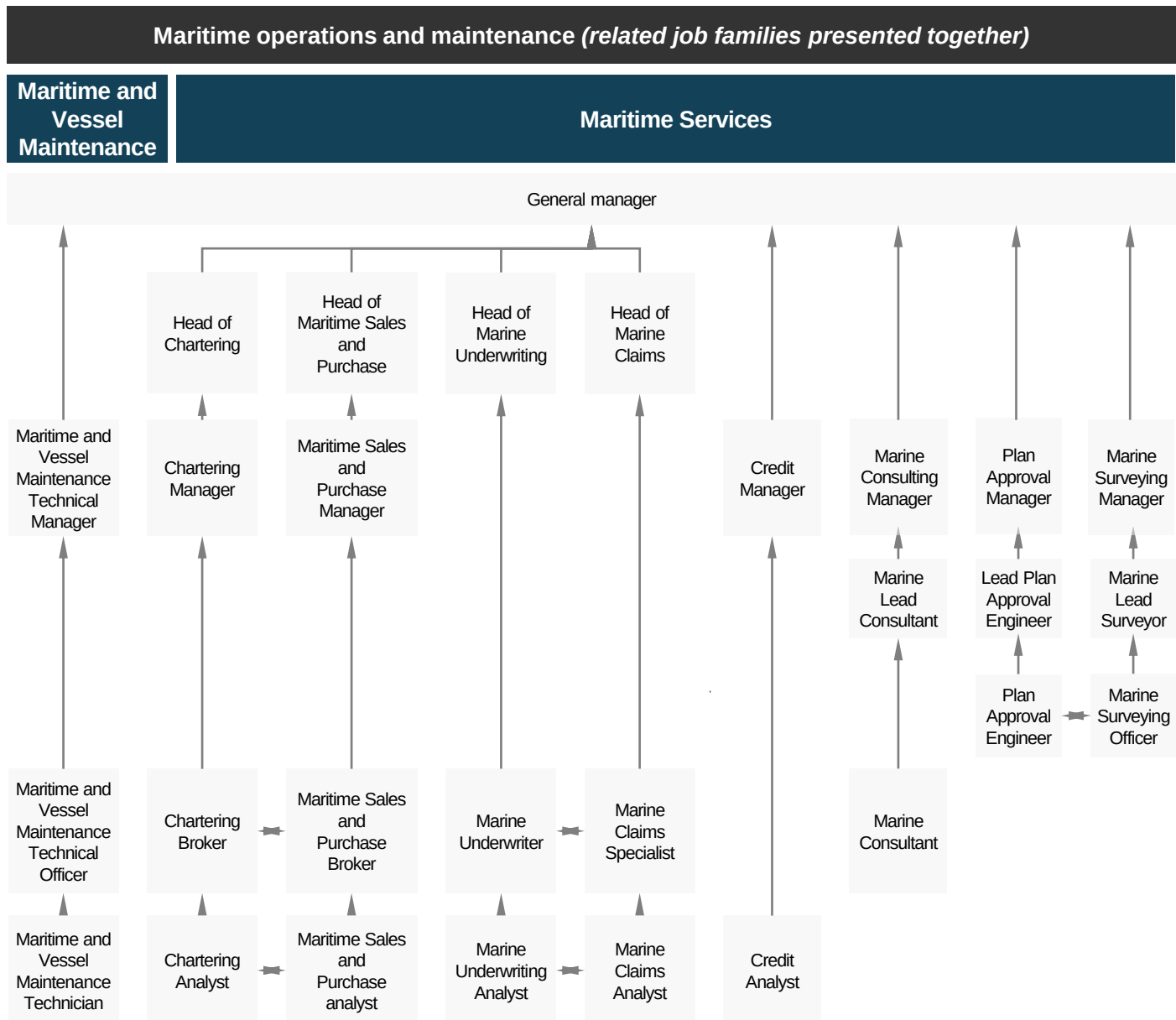
Key tasks/activities include:

- Undertaking chartering brokerage activities
- Carrying out marine underwriting
- Handling claims and surveying
- Providing maritime legal and insurance services
- Executing cargo loading operations
- Monitoring maritime traffic and communications

The family **has 26 key job roles**:

General manager, Head of chartering, Chartering manager, Chartering broker, Chartering analyst, Head of maritime sales and purchase, Maritime sales and purchase manager, Maritime sales and purchase broker, Maritime sales and purchase analyst, Head of marine underwriting, Marine underwriter, Marine underwriting analyst, Head of marine claims, Marine claims specialist, Marine claims analyst, Credit manager, Credit analyst, Marine consulting manager, Marine lead consultant, Marine consultant, Plan approval manager, Lead plan approval engineer, Plan approval engineer, Marine surveying manager, Marine lead surveyor, Marine surveying officer

Job family profiles and career pathway maps: (12) Maritime and Vessel Maintenance (13) Maritime Services



Relevant USOC unit-tier codes for the job family: 1120 Managing directors and chief executives, 3151 Ships' engineers, 7233 Agricultural and industrial machinery mechanics and repairers, 1321 Manufacturing managers, 1211 Finance managers, 3115 Mechanical engineering technicians, 1324 Supply, distribution and related managers, 3321 Insurance representatives, 3322 Commercial sales representatives, 3324 Trade brokers

Job family profiles and career pathway maps: (12) Maritime and Vessel Maintenance (13) Maritime Services

Skills requirements

Roles under Maritime and Vessel Maintenance require specialized skills in maritime maintenance. The main skills include ships engineering, preventive maintenance, port equipment maintenance, etc. While the roles under Maritime Services are more focused operations. The main skills include container loading and unloading administration, engine room operations, marine incident and accident investigations, etc.

Full list of key skills is present in the following tables (mentioned below) providing a detailed view on job families' skills needs.

Key relevant skills: Maritime and Vessel Maintenance

- | | |
|--|--|
| • Autonomous systems engineering and maintenance | • Ship maintenance and repair |
| • Ships engineering | • Vessel handling |
| • Preventive maintenance | • Vessel repair and maintenance |
| • Mobile equipment - prime mover | • Workplace safety and health for crane operations |
| • defensive driving | • Workplace health, safety and security procedures |
| • Port equipment maintenance | • Single stack medium forklift operations |

Key relevant skills: Maritime Services

- | | |
|---|--|
| • Cargo valuation and certification | • Marine survey reporting |
| • Container loading and unloading administration | • Marine underwriting |
| • Deck assistance and electronic data interchange | • Chartering |
| • Engine room operations | • Ship safety management systems audit |
| • Marine incident and accident investigations | |
| • Marine insurance claims handling | |

Job family profiles and career pathway maps: (12) Maritime and Vessel Maintenance (13) Maritime Services

Relevant certifications/ licenses (non-exhaustive)

Both the job families are rooted in strong engineering expertise with a **significant presence of e-logistics, additional product services, and logistic solutioning** for the roles. **Membership with the Saudi Council of Engineers** is mandatory for all practicing engineers and technicians in the Kingdom and the requirement is therefore also applicable to the engineering roles under these job families. Additionally, **Maritime license issued by the Saudi Ports Authority** is mandatory for performing activities related to maritime operations and services. Other than the license and the membership, there are various highly acknowledged certifications for these roles (mentioned below) that are prevalent in the Kingdom.

Certifications: Maritime and Vessel Maintenance

- Shipbuilding/ repair courses at National Maritime Academy, KSA
- Fundamentals of Engineering Certificate (FE) by National Council of Examiners for Engineering and Surveying

Certifications: Maritime Services

- ICS (Institute of Chartered Shipbrokers) Qualification
- Chartered Insurance Institute (CII) Qualifications
- International Certificate in Logistics and Transport (CILT)

Job family profiles and career pathway maps: (14) Port Operations

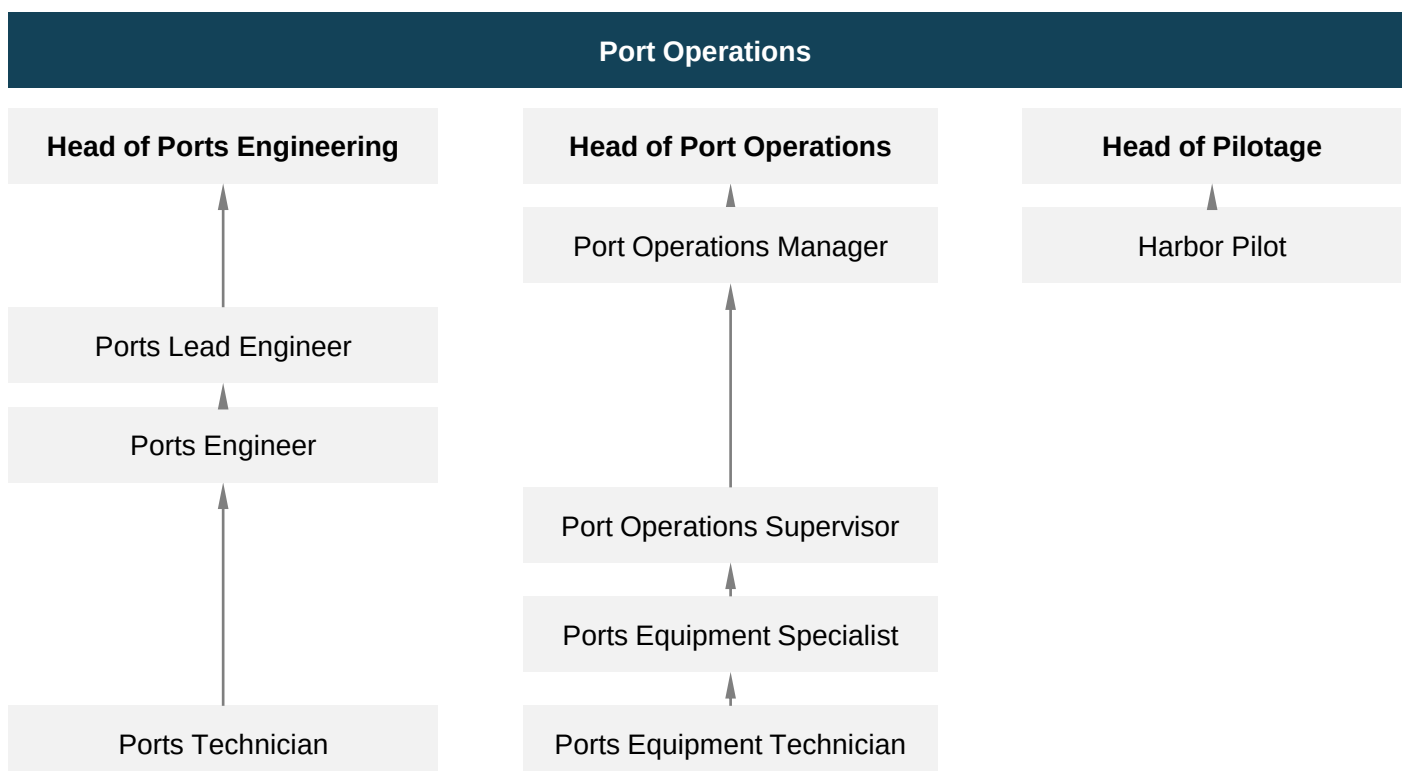
Job family overview

The job family clusters roles that focus on overseeing functioning of **port facilities, ensuring efficient ship management** from arrival to departure, **and coordinating services** to facilitate the handling of vessels and cargo

Key tasks/activities include:

- Coordinating and Documenting
- Stevedoring
- Harbor piloting
- Providing container service yard operations
- Dredging
- Bunkering
- Undertaking terminal operations

The family has **11 key job roles**: Head of ports engineering, Head of port operations, Head of pilotage, Port operations manager, Harbor pilot, Ports lead engineer, Ports operations supervisor, Ports Engineer, Ports equipment specialist, Ports equipment technician, Ports technician.



Relevant USOC unit-tier codes for the job family: 1324 Supply, distribution and related managers, 3152 Ships' deck officers and pilots, 8350 Ships' deck crews and related workers, 3115 Mechanical engineering technicians, 3115 Mechanical engineering technicians

Job family profiles and career pathway maps:

(14) Port Operations

Skills requirements

The listed job roles require specialized skills relate to port operations and management. The main skills include automated logistics and port infrastructure management, cross docking, container loading and unloading administration, goods handling and moving, occupational machines operations, etc.

Full list of key skills is present in the following table providing a detailed view on job family skills needs.

Key relevant skills

- Automated logistics and port infrastructure management
- Cross docking
- Container loading and unloading administration
- Goods handling and moving
- Occupational machines operations
- Port and terminal operations management
- Port call planning
- Warehouse operations
- Ship mooring, anchoring and securing operations
- Shipment classification and clearance

Relevant certifications/ licenses (non-exhaustive)

The job family roles require strong knowledge of **port operations and maintenance**. The job family roles require port engineers and technicians for the jobs. In the Kingdom, practicing engineers and technicians are required by law to hold **membership with the Saudi Council of Engineers (SCE)**, ensuring that professionals meet national standards and adhere to engineering regulations. Additionally, **Maritime license issued by Saudi Ports Authority** is required for port operation activities. Other than the membership and license, there are various highly acknowledged certifications for these roles (mentioned below) that are prevalent in the Kingdom.

Certifications

- Fundamentals of Engineering Certificate (FE) by National Council of Examiners for Engineering and Surveying
- Port Operations Course by National Maritime Academy, KSA
- International Certificate in Logistics and Transport (CILT)

Job family profiles and career pathway maps:

(15) Freight Forwarding Operations

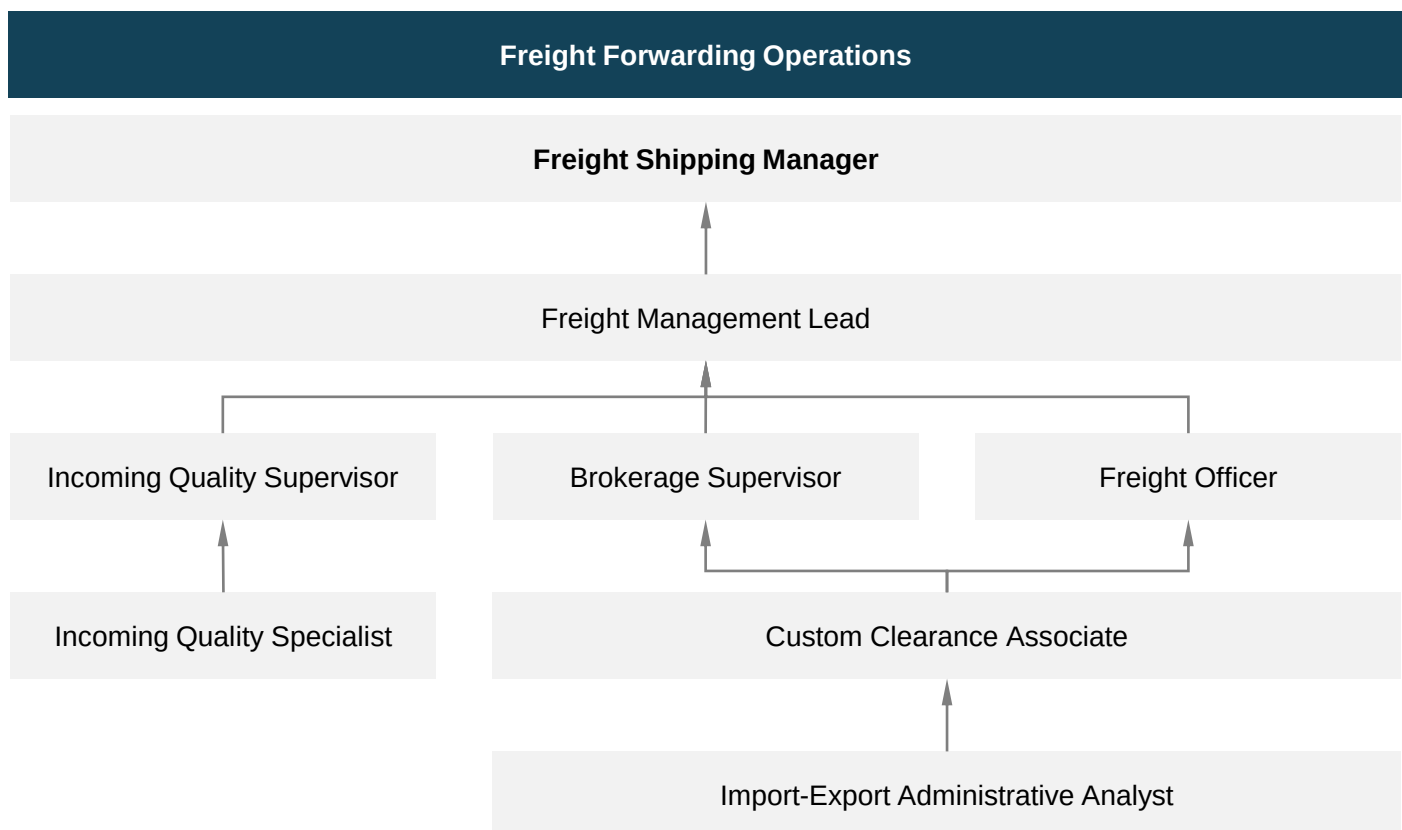
Job family overview

The job family clusters roles that focus on performing activities required **for movement of goods from one location to another safely**, and in compliance with all regulatory requirements, typically through multiple modes of transportation

Key tasks/activities include:

- Providing customs brokerage and clearing facilities
- Coordinating import/ export activities
- Managing documents
- Planning shipments and optimizing routes
- Providing risk management services

The family has **8 key job roles**: Freight shipping manager, Freight management lead, Incoming quality supervisor, Brokerage supervisor, Freight officer, Incoming quality specialist, Custom clearance associate, Import-export administrative analyst



Relevant USOC unit-tier codes for the job family: 3324 Trade brokers, 3331 Clearing and forwarding agents, 2421 Management and organization analysts, 1213 Policy and planning managers

Job family profiles and career pathway maps:

(15) Freight Forwarding Operations

Skills requirements

The listed job roles require specialized skills and knowledge related to freight forwarding. The main skills include customs clearance and procedures, freight insurance administration, digital supply chain development and maintenance, freight knowledge and documentation, etc.

Full list of key skills is present in the following table providing a detailed view on job family skills needs.

Key relevant skills

- | | |
|--|--|
| • Customs clearance and procedures | • Goods handling and moving |
| • Freight insurance administration | • Import and export documentation administration |
| • Digital supply chain development and maintenance | • Logistics process quality management |
| • Freight Knowledge and Documentation | • Port and terminal operations management |
| • Good distribution practices | • Risk analytics |

Relevant certifications/ licenses (non-exhaustive)

The job family roles require **freight forwarding officers, clearance brokers and quality inspectors** for the jobs. In the Kingdom, custom brokers are legally required to hold a valid **Customs Brokers license issued by the Zakat, Tax, and Customs Authority** to practice the profession. In addition to this, there is a prevalent certification for the listed job roles (mentioned below) in the Kingdom.

Certifications

- FIATA (International Federation of Freight Forwarders Associations) certificate/ Diploma in Freight Forwarding

Job family profiles and career pathway maps:

(16) Purchasing and Procurement Operations (17) Special Product Handling

Job family overview

Purchasing and Procurement and Special Product Handling job families have closely related job roles contributing to the **timely sourcing, cost-efficient management, and safe handling of goods and specialized products**, ensuring quality, compliance and supply chain operations. Given the synergies, the **two families are presented on a single career pathway map**

Purchasing and Procurement Operations

The job family clusters roles that focus on **managing procurement of goods and services** by ensuring timely sourcing, maintaining supplier relationships, and negotiating contracts for optimizing cost efficiency

Key tasks/activities include:

- Analysing procurement needs
- Budgeting
- Tracking inventory
- Managing purchase orders
- Negotiating and onboarding suppliers
- Liaising with vendors
- Ensuring product quality
- Conducting Supplier audits

The family **has 4 key job roles:**

Head of procurement, Procurement manager, Procurement associate, Procurement analyst

Special Product Handling

The job family clusters roles that focus on **handling of special or perishable products in a safe and efficient manner** with careful transportation and storage. Products include temperature-sensitive items such as agriculture products, pharmaceuticals and healthcare, liquids, mining, etc.

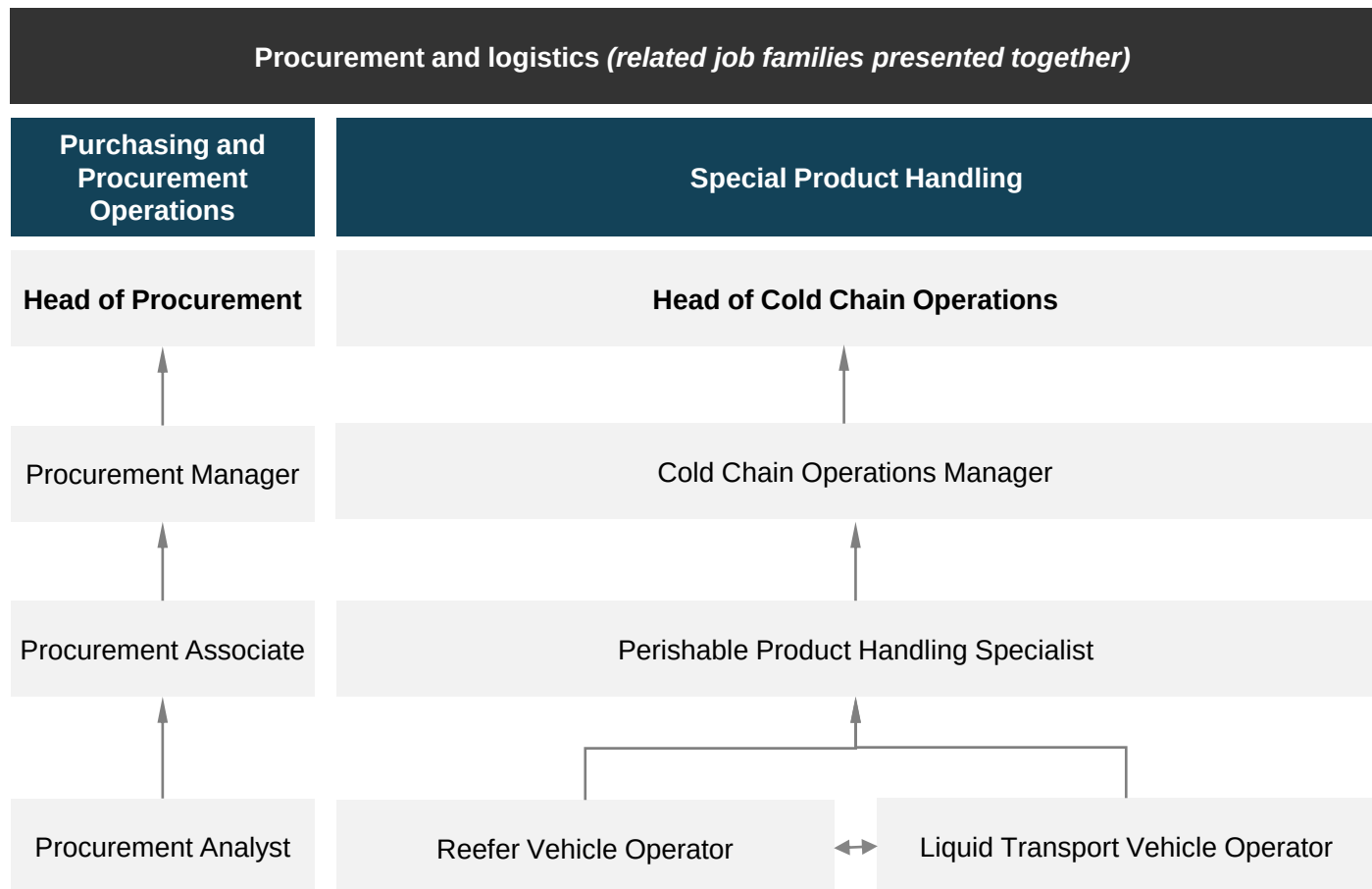
Key tasks/activities include:

- Deploying specialized equipment
- Packaging, labeling, or storage conditions to ensure the integrity, safety, and quality of the product
- Ensuring regulatory compliance
- Managing quality assurance processes
- Conducting inspections for quality control

The family **has 5 key job roles:**

Head of cold chain operations, Cold chain operations manager, Perishable product handling specialist, Reefer vehicle operator, Liquid transport vehicle operator

Job family profiles and career pathway maps: (16) Purchasing and Procurement Operations (17) Special Product Handling



Relevant USOC unit-tier codes for the job family: 1324 Supply, distribution and related managers, 3323 Buyers, 1324 Supply, distribution and related managers, 8332 Heavy truck and lorry drivers

Skills requirements

The listed job roles require specialized skills related to procurement of goods and management of transportation for special products. The main skills for Purchasing and Procurement Operations include contract development and management, vendor management, etc. The main skills for Special Product Handling include automated logistics and port infrastructure management, automated warehousing systems integration, cold chain operations administration, etc.

Full list of key skills is present in the following tables (see next page) providing a detailed view on job families' skills needs.

Job family profiles and career pathway maps:

(16) Purchasing and Procurement Operations (17) Special Product Handling

Key relevant skills: Purchasing and Procurement Operations

- Contract Development and Management
- Vendor management
- Financial planning and budget management
- Material inspection
- On-demand logistics coordination
- Inventory management
- Import and export documentation administration
- Freight Knowledge and Documentation
- Procurement planning
- Purchasing management
- Quality assurance

Key relevant skills: Special Product Handling

- Automated logistics and port infrastructure management
- Automated warehousing systems integration
- Cold chain operations administration
- Consignments processing and tracking
- Container loading and unloading administration
- Hazardous materials identification system (HMIS) administration
- Logistics risk management
- On-demand logistics coordination
- Product and goods inspection
- Specialized transport operations

Relevant certifications/ licenses (non-exhaustive)

The job family roles require expertise in **of procurement, supply chain, and goods handling and transportation**. In the Kingdom, drivers for trucks and heavy vehicles are legally required to hold a valid Heavy Vehicle Operations license issued by Transport General Authority. Therefore, the license is also applicable to the vehicle operators in the Special Product Handling job family. Other than this, there are various highly acknowledged certifications for these roles (mentioned below) that are prevalent in the Kingdom.

Certifications: Purchasing and Procurement Operations

- Certified International Procurement Professional (CIPP)
- Certified International Procurement Manager (CIPM)
- Certified International Supply Chain Professional (CISCP)

Certifications: Special Product Handling

- Certified Supply Chain Professional (CSCP) Certification
- Special cargo handling diploma offered by International Air Transport Association

Job family profiles and career pathway maps:

(18) Analysis and Reporting (19) Demand Forecasting (20) Warehouse and Fulfilment Center Operations

Job family overview

Analysis and Reporting, Demand Forecasting, and Warehouse and Fulfillment Center Operations job families have closely related roles **that together contribute to the inventory management function**. Given the synergies, the **three families are presented on a single career pathway map**.

Analysis and Reporting

The job family clusters roles that focus on using technology and data to monitor, analyze, and summarize key metrics, **delivering insights on performance and trends in operations and inventory**.

Key tasks/activities include:

- Aggregating and interpreting data
- Developing reporting frameworks
- Conducting trend analysis
- Building dashboards
- Reporting and benchmarking KPIs
- Performing scenario planning and analysis

The family **has 5 key job roles**: Warehouse operations manager, Warehouse operations lead, Inventory supervisor, Inventory coordination specialist, Data feeding analyst

Demand Forecasting

The job family clusters roles that deal with **projecting future product requirements to inform inventory management** and ensure smooth retrieval and dispatch in warehouses.

Key tasks/activities include:

- Analyzing historical warehousing data
- Interpreting market trends
- Assessing consumer behaviors to forecast demand accurately

The family **has 5 key job roles**: Warehouse operations manager, Warehouse operations lead, Inventory supervisor, Inventory coordination specialist, Inventory analyst

Warehouse and Fulfilment Center Operations

The job family clusters roles that **oversee daily processes within facilities to ensure efficient storage and handling of goods** along with inventory management.

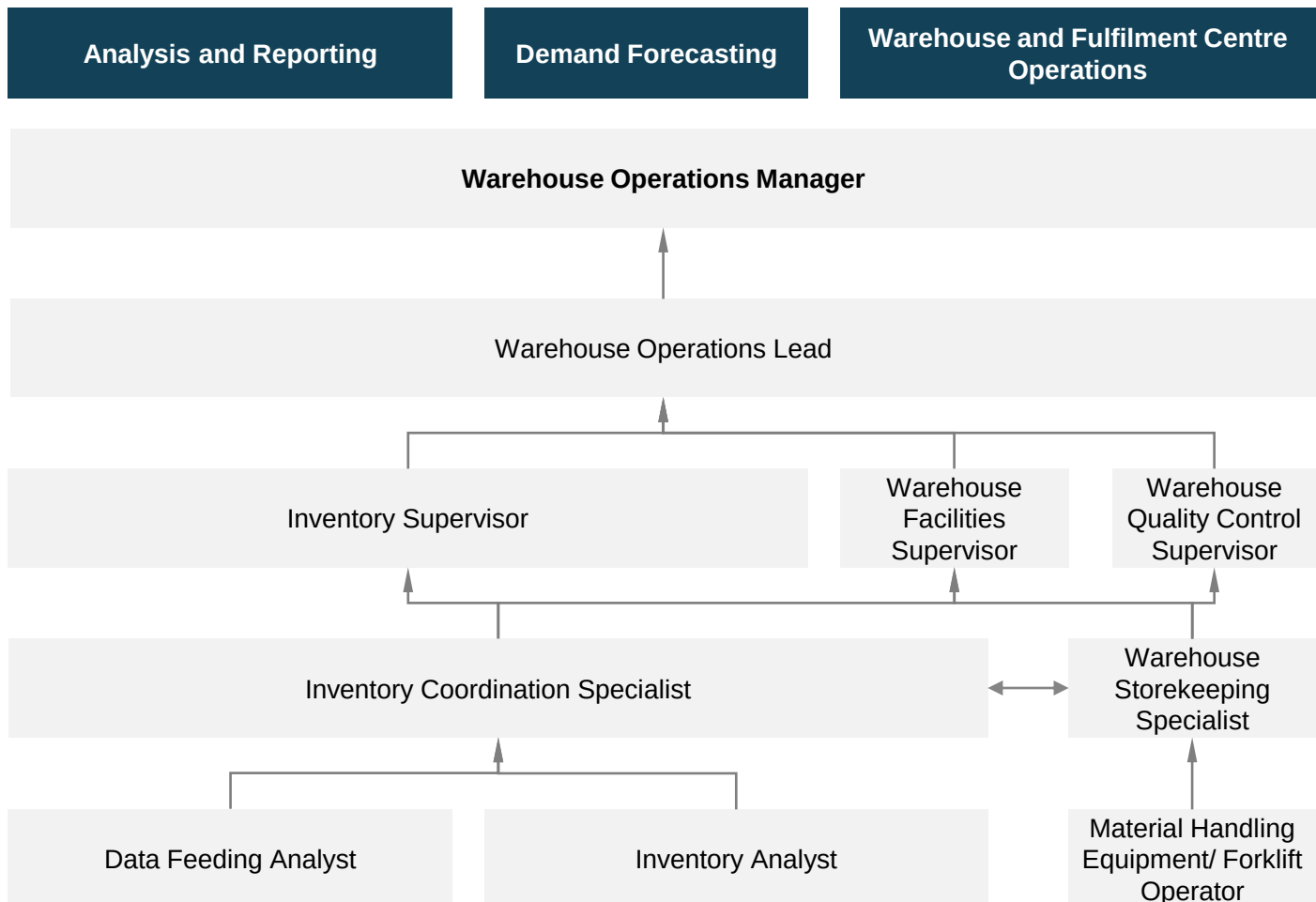
Key tasks/activities include:

- Managing inventory
- Inventory audits
- Processing orders
- Pick and pack services
- Shelving and organizing
- Bundling and assembly
- Labelling and barcoding
- Packaging
- Scrap management
- Reverse logistics management

The family **has 8 key job roles**: Warehouse operations manager, Warehouse operations lead, Warehouse facilities supervisor, Warehouse quality control supervisor, Warehouse storekeeping specialist, Inventory coordination specialist, Inventory analyst, Material handling equipment/ forklift operator

Job family profiles and career pathway maps: (18) Analysis and Reporting (19) Demand Forecasting (20) Warehouse and Fulfilment Center Operations

Warehouse operations and inventory management (related job families presented together)



Relevant USOC unit-tier codes for the job family: 1213 Policy and planning managers, 1324 Supply, distribution and related managers, 2411 Accountants, 3313 Accounting associate professionals, 4132 Data entry clerks, 8344 Lifting truck operators, 4321 Stock clerks, 9334 Shelf fillers

Job family profiles and career pathway maps:

(18) Analysis and Reporting (19) Demand Forecasting

(20) Warehouse and Fulfilment Center Operations

Skills requirements

The listed roles require specific skills related to inventory data processing, insights delivery, and stock management. Roles under Analysis and Reporting require specialized skills in data analysis and reporting. The main skills include data analytics, forecast and trend analysis, etc. Similarly, roles under Demand Forecasting require specialized skills in trend forecasting and analysis. The main skills include business plan development and implementation, demand analysis, etc. The roles under Warehouse and Fulfilment Center Operations require specialized skills in warehouse operations. The main skills include cargo warehousing operations, forklift operations, package and order picking, etc.

Full list of key skills is present in the following tables (given below) providing a detailed view on job families' skills needs.

Key relevant skills: Analysis and Reporting

- Consignments processing and tracking
- Data analytics
- Forecast and trend analysis
- Inventory management
- Resource allocation and operations streamlining
- Supply chain solutioning/modelling/ planning/ strategizing
- Supply chain traceability analysis
- Technology integration and application
- Transport planning and coordination
- Warehouse inventory control/audit

Key relevant skills: Demand Forecasting

- Business plan development and implementation
- Demand analysis
- Financial planning and budget management
- Logistics process quality management
- On-demand logistics coordination
- Organizational strategy and policy realization
- Pricing strategy
- Warehouse inventory control/audit
- Warehouse performance measurement
- Supply chain traceability analysis

Key relevant skills: Warehouse and Fulfilment Center Operations

- Cargo warehousing operations
- Forklift operations
- Package and order picking
- Order fulfilment administration
- Warehouse administration
- Warehouse facility management
- Warehouse goods handling
- Warehouse inventory control and audit
- Warehouse management system administration
- Warehouse operations
- Warehousing/cargo-related occupational health & safety mgt

Job family profiles and career pathway maps: (18) Analysis and Reporting (19) Demand Forecasting (20) Warehouse and Fulfilment Center Operations

Relevant certifications/ licenses (non-exhaustive)

The job family roles require strong expertise in **inventory analytics and management**. There are various highly acknowledged certifications for these roles (mentioned below) that are prevalent in the Kingdom.

Certifications: Analysis and Reporting

- Certified Analytics Professional (CAP) credential
- Associate Big Data Analyst (ABDA)

Certifications: Demand Forecasting

- Certification of Production and Inventory Management (CPIM) offered by ASCM
- Certified Professional Forecaster (CPF) offered by ASCM

Certifications: Warehouse and Fulfilment Center Operations

- Certified Warehousing Operations Expert (CWOE) by The Supply Chain Institute

Job family profiles and career pathway maps: (21) E-logistics (22) Additional Product Service (23) Logistics Solutioning

Job family overview

E-logistics, Additional Product Services, and Logistics Solutioning have closely related job roles that together contribute to **overall logistics process and services facilitation**. Given the synergies, the **three families are presented on a single career pathway map**.

E-logistics

The job family clusters roles that deal with managing **e-logistics encompassing the entire online order lifecycle**, from initial fulfillment to the seamless handling of returns and exchanges, specifically in e-commerce.

Key tasks/ activities include:

- Managing E-commerce platform
- Processing Order/exchange request
- Managing order fulfillment and delivery coordination
- Managing reverse logistics
- Managing customer relationship
- Managing suppliers

The family **has 4 key job roles**: E-commerce director, E-commerce manager, E-commerce associate, E-commerce analyst

Additional Product Services

The job family clusters roles that deal with enhancing value for customers through **product customization, including final assembly, light manufacturing, and printing**, as well as remanufacturing and refurbishing of returned products.

Key tasks/ activities include:

- Managing product customization
- Overseeing final assembly operations
- Coordinating small-scale manufacturing
- Executing product refurbishing
- Performing quality repairs
- Conducting product finishing activities
- Managing inventory

The family **has 4 key job roles**: Additional product services manager, Lead production engineer for additional product services, Production engineer for additional product services, Technician for additional product services

Logistics Solutioning

The job family clusters roles that deal with creating **tailored strategies and systems to solve specific logistics or supply chain challenges**, optimizing efficiency, costs, and performance.

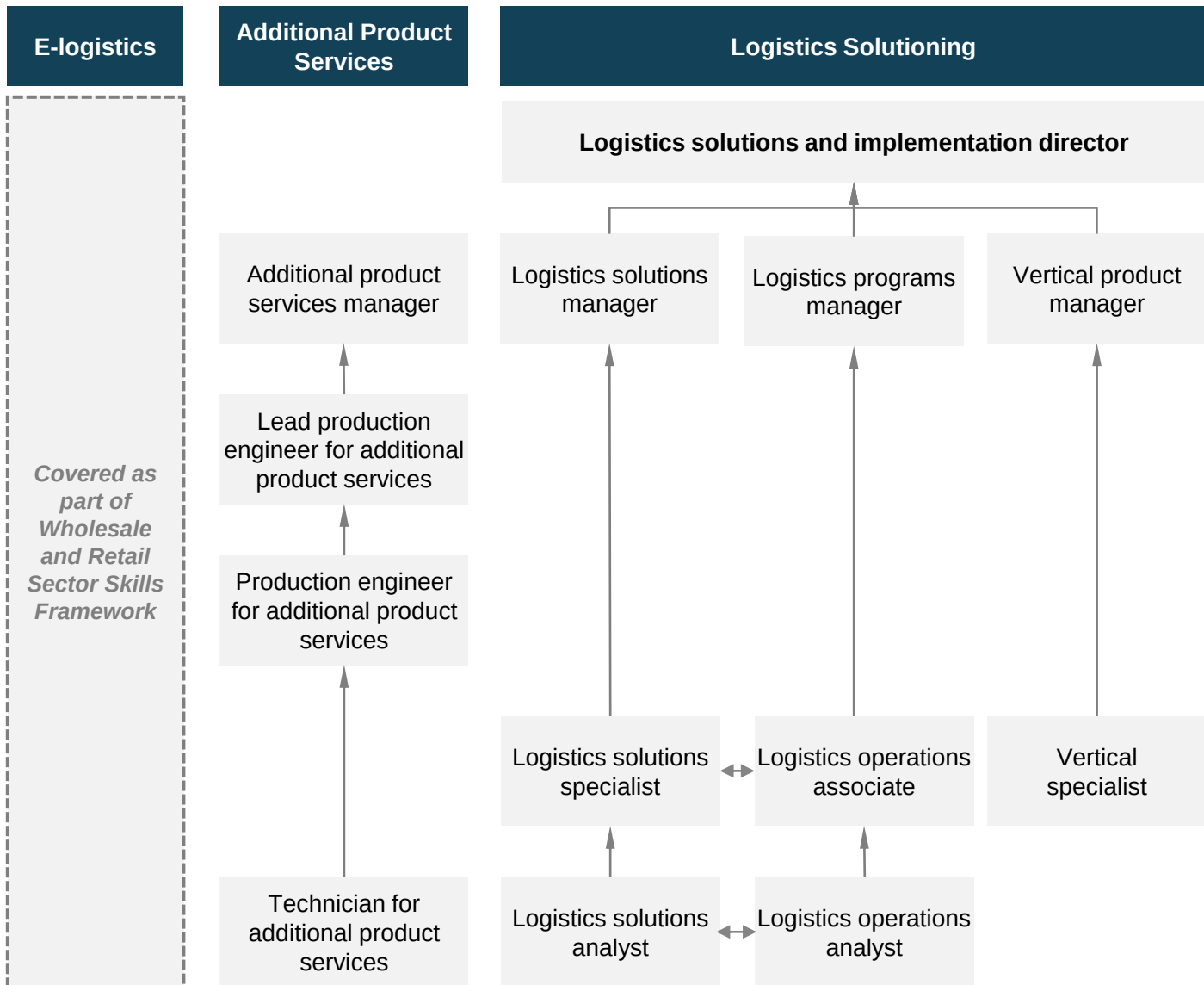
Key tasks/ activities include:

- Optimizing inventory management
- Identifying suitable transportation modes
- Delivering integrated fourth-party logistics solutions
- Managing warehouse operations
- Designing logistics networks and selecting carriers
- Managing supplier relationships

The family **has 9 key job roles**: Logistics solutions and implementation director, Logistics solutions manager, Logistics programs manager, Vertical product manager, Vertical specialist, Logistics solution specialist, Logistics operations associate, Logistics solution analyst, Logistics operations analyst

Job family profiles and career pathway maps: (21) E-logistics (22) Additional Product Service (23) Logistics Solutioning

Logistics operations and solutioning (*related job families presented together*)



Relevant USOC unit-tier codes for the job family: 1420 Retail and wholesale trade managers, 1321 Manufacturing managers, 8211 Mechanical machinery assemblers, 2141 Industrial and production engineers, 1321 Manufacturing managers, 8211 Mechanical machinery assemblers, 2141 Industrial and production engineers, 1223 Research and development managers, 1324 Supply, distribution and related managers.

Job family profiles and career pathway maps: (21) E-logistics (22) Additional Product Service (23) Logistics Solutioning

Skills requirements

Roles under E-logistics require specialized skills in logistics management. The main skills for E-logistics include cargo tracking systems administration, customer experience management, etc. While the roles under Additional product services require specialized skills in product services and management. The main skills include goods packaging and labelling, process improvement and optimization, product and goods inspection etc. Logistics solutioning requires skills mainly related to logistics process improvement. The main skills for Logistics solutioning include logistics support, courier delivery, etc.

Full list of key skills is present in the following tables providing a detailed view on job families' skills needs.

Key relevant skills: E-logistics

- | | |
|---|---|
| <ul style="list-style-type: none"> • Cargo tracking systems administration • Customer experience management • E-logistics IT solutioning • Good distribution practices • Logistics solutions sales | <ul style="list-style-type: none"> • Logistics solution product/project management • Order fulfilment administration • Transport management system administration • Vendor management • Warehouse management system administration |
|---|---|

Key relevant skills: Additional Product Services

- | | |
|---|--|
| <ul style="list-style-type: none"> • Customer experience management • Goods packaging and labelling • Logistics Design thinking • Logistics operations research and planning • Omnichannel supply continuity | <ul style="list-style-type: none"> • Process improvement and optimization • Product and goods inspection • Risk Management and Administration • Technology integration and application |
|---|--|

Key relevant skills: Logistics Solutioning

- | | |
|---|---|
| <ul style="list-style-type: none"> • Analysis and evaluation in defense logistics • Courier delivery operations • Distribution networks development • Event logistics administration • Loading crane operations • Logistics support | <ul style="list-style-type: none"> • Logistics risk and technology management • Strategic logistics planning • Network planning and operations • Occupational machines operations |
|---|---|

Job family profiles and career pathway maps: (21) E-logistics (22) Additional Product Service (23) Logistics Solutioning

Relevant certifications/ licenses (non-exhaustive)

The Additional Product Services and Logistics Solutioning job families are rooted in strong engineering expertise with a **significant presence of product and logistics engineers and technicians** for the roles, while the E-logistics job family requires e-commerce platform and logistics management specialists. **Membership with the Saudi Council of Engineers** is mandatory for all practicing engineers and technicians in the Kingdom and the requirement is therefore also applicable to the engineering roles under the job families. In addition to this, there are various highly acknowledged certifications for the listed roles (mentioned below) that are prevalent in the Kingdom.

Certifications: E-Logistics

- Certified Six Sigma Black Belt
- Certified International Supply Chain Manager (CISCM)
- Introductory Certificate in Logistics and Transport (CILT)

Certifications: Additional Product Services

- Fundamentals of Engineering Certificate (FE) by National Council of Examiners for Engineering and Surveying

Certifications: Logistics Solutioning

- Fundamentals of Engineering Certificate (FE) by National Council of Examiners for Engineering and Surveying
- Certified International Supply Chain Professional (CISCP)
- Introductory Certificate in Logistics and Transport (CILT)

Job family profiles and career pathway maps: (24) Sales and Customer Service

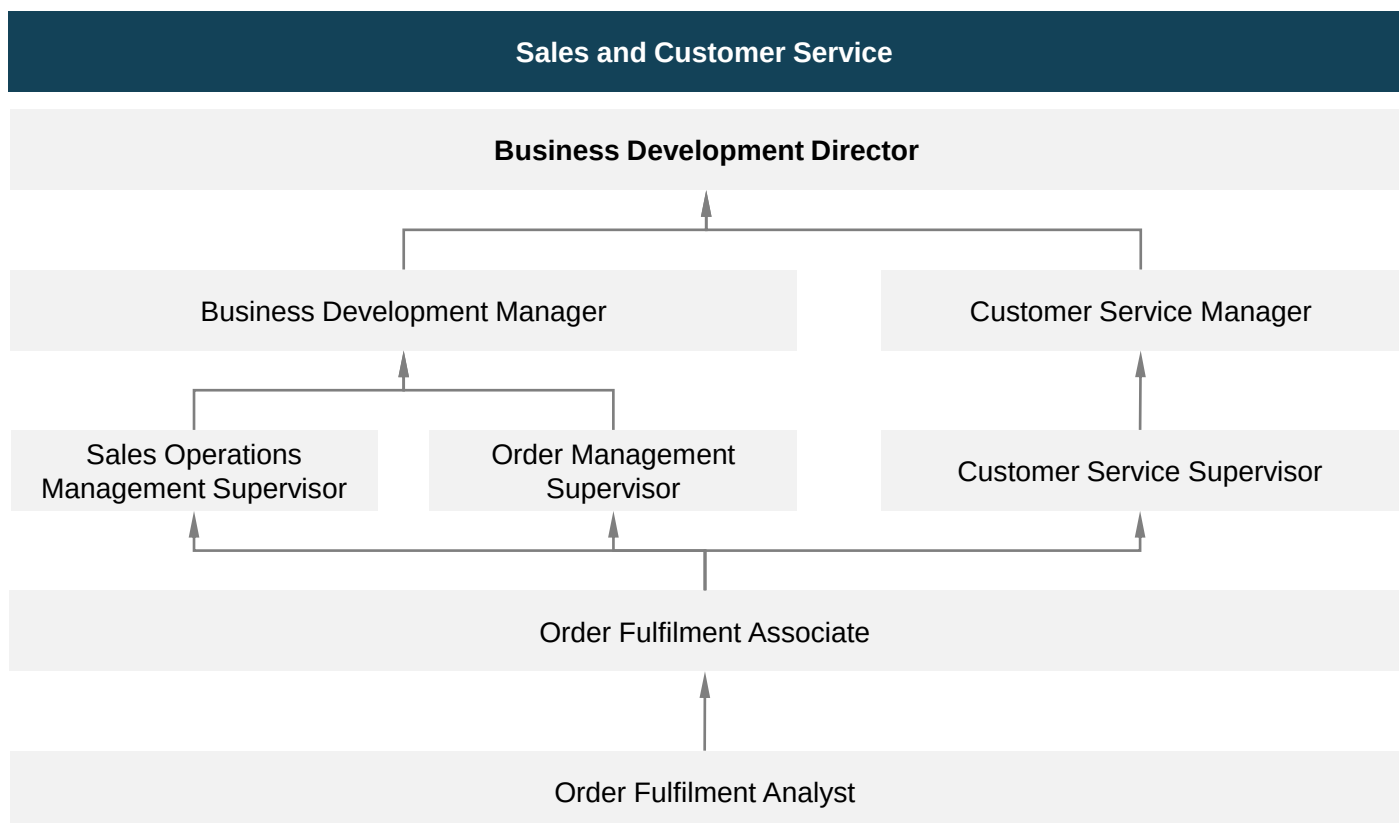
Job family overview

The job family clusters roles that focus on building **strong relationships with clients and selling services** to customers and providing them assistance before, during, and after the purchase.

Key tasks/activities include:

- Driving business development
- Strategizing sales initiatives
- Providing after-sales support and guidance
- Developing marketing strategies

The family has **8 key job roles**: Business development director, Business development manager, Customer service manager, Sales operations management supervisor, Order management supervisor, Customer service supervisor, Order fulfilment associate, Order fulfilment analyst



Relevant USOC unit-tier codes for the job family: 1221 Sales and marketing managers, 2431 Advertising and marketing professionals, 3322 Commercial sales representatives, 5242 Sales demonstrators, 4222 Contact center information clerks, 4225 Enquiry clerks

Job family profiles and career pathway maps: (24) Sales and Customer Service

Skills requirements

The listed job roles require skills mainly related to sales and customer care. The main skills include customer experience management, customer service delivery, customer relationship management, market research and analysis, pre-sales and post-sales activities, etc.

Full list of key skills is present in the following table providing a detailed view on job family skills needs.

Key relevant skills

- | | |
|---------------------------------------|--|
| • Customer experience management | • Logistics solutions sales |
| • Customer service delivery | • Order fulfilment administration |
| • Customer relationship management | • Process improvement and optimization |
| • Market research and analysis | • Sales planning |
| • Pre-sales and post-sales activities | • Sales strategy |

Relevant certifications/ licenses (non-exhaustive)

The job family roles require strong expertise in **sales and customer service**. The prevalent certifications for the listed job roles in the Kingdom are listed below.

Certifications

- CBP™ Sales Certification
- CBP™ Customer Service Certification

Job family profiles and career pathway maps:

(25) Network Planning (26) Logistics Process Improvement

Job family overview

Network Planning and Logistics Process Improvement job families have closely related job roles contributing to **optimization and improvement of logistics and supply chain**. Given the synergies, the **two families are presented on a single career pathway map**.

Network Planning

The job family clusters roles that deal with **designing and optimizing physical infrastructure** and routes involved in moving cargo and passengers for reduced travel/ delivery time.

Key tasks/activities include:

- Analysing and planning routes
- Planning capacity
- Simulating route scenarios through network modelling
- Designing Infra and optimizing for any adjustments
- Implementing tech solutions for route optimization

The family **has 4 key job roles:**

Director network planning, Network planning manager, Network planning supervisor, Network planning analyst.

Logistics Process Improvement

The job family clusters roles that deal with enhancement of logistics processes by **optimizations and implementation of IT solutions** such as automated inventory management systems, real-time tracking, etc.

Key tasks/activities include:

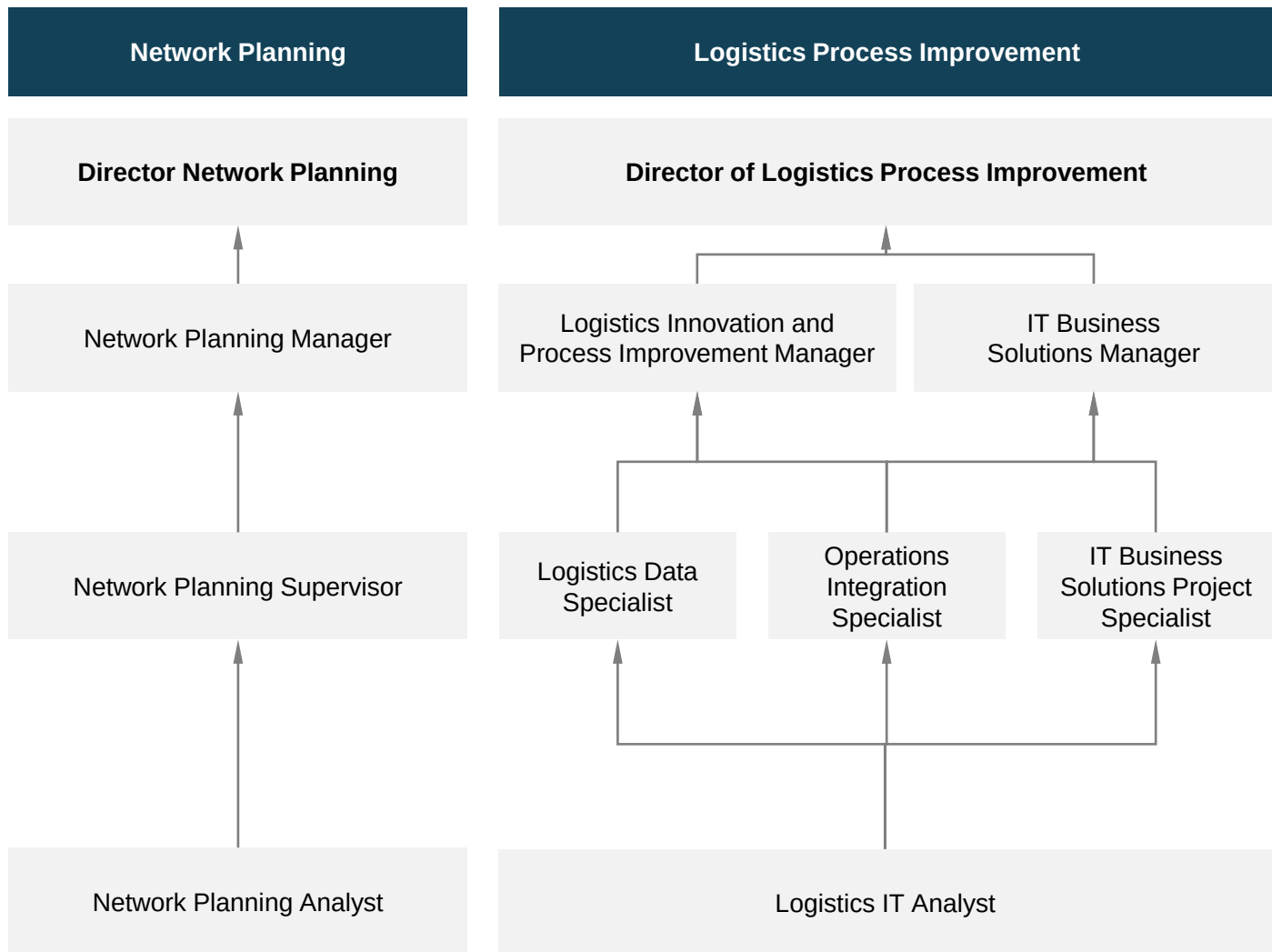
- Collecting data of logistics process
- Implementing system-wide data integration across supply chain platforms
- Conducting detailed analysis of inventory turnover and demand variability
- Mapping out current logistics workflows to identify inefficiencies and implementing optimal solutions
- Deploying IT and automation solutions

The family **has 7 key job roles:**

Director of logistics process improvement, Logistics innovation and process improvement manager, IT business solutions manager, Logistics data specialist, Operations integration specialist, IT business solutions project specialist, Logistics IT analyst.

Job family profiles and career pathway maps: (25) Network Planning (26) Logistics Process Improvement

Logistics infrastructure and process enhancement (related job families presented together)



Relevant USOC unit-tier codes for the job family: 1213 Policy and planning managers, 2142 Civil engineers 2522 3112 Civil engineering technicians, 1323 Construction managers, 1324 Supply, distribution and related managers, 1321 Manufacturing managers, 1324 Supply, distribution and related managers, 1330 Information and communications technology service managers, 3139 Process control technicians not elsewhere classified

Job family profiles and career pathway maps: (25) Network Planning (26) Logistics Process Improvement

Skills requirements

The job roles under Network Planning job family require skills related to logistics infra design and development. The main skills include customer transport requirements, demand analysis, strategy development, etc. While job roles under Logistics Process Improvement job family require skills related to logistics automation and process optimization. The main skills include automated logistics and port infrastructure, cargo tracking systems administration, etc.

Full list of key skills is present in the following tables (given below) providing a detailed view on job families' skills needs.

Key relevant skills: Network Planning

- | | |
|-------------------------------------|--|
| • Customer transport requirements | • Strategy development and planning |
| • Demand analysis | • Transportation route and schedule planning |
| • Navigation & route identification | • Transportation route optimization |
| • Network planning and operations | |
| • Navigation & route identification | |
| • Pricing strategy | |
| • Route planning and operations | |

Key relevant skills: Logistics Process Improvement

- | | |
|--|--|
| • Automated logistics and port infrastructure management | Cargo tracking systems administration |
| • Automated warehousing systems integration | Process improvement and optimization |
| • Automation Design | Technology integration and application |
| • Autonomous logistics Design and application | Robots in logistics |
| • Autonomous systems engineering and maintenance | Logistics process quality management |

Relevant certifications/ licenses (non-exhaustive)

The job families are rooted in strong engineering expertise with a **significant presence of network planning and logistics IT solutions engineers and specialists** for the roles. **Membership with the Saudi Council of Engineers** is mandatory for all practicing engineers and technicians in the Kingdom and the requirement is therefore also applicable to the engineering roles under these job families. In addition to this, there are various highly acknowledged certifications for these roles (mentioned on the next page) that are prevalent in the Kingdom.

Job family profiles and career pathway maps: (25) Network Planning (26) Logistics Process Improvement

Certifications: Network Planning

- Fundamentals of Engineering Certificate (FE) by National Council of Examiners for Engineering and Surveying
- Certifications through Association for Supply Chain Management (ASCM)
- International Certificate in Logistics and Transport (CILT)

Certifications: Logistics Process Improvement

- Fundamentals of Engineering Certificate (FE) by National Council of Examiners for Engineering and Surveying
- International Certificate in Logistics and Transport (CILT)
- Certified International Supply Chain Professional (CISCP)
- Certified International Supply Chain Manager (CISCM)
- Certified Six Sigma Black Belt

Job family profiles and career pathway maps:

(27) Engineering and Technical Service (28) Advanced Maintenance

Job family overview

Engineering and Technical Service and Advanced Maintenance job families have closely related job roles contributing to the **overall equipment and systems maintenance**. Given the synergies, the **two families are presented on a single career pathway map**.

Engineering and Technical Service

The job family clusters roles that deal with **configuration, maintenance, and enhancement of critical material handling equipment and logistics systems** essential for warehouse operations.

Key tasks/activities include:

- Calibrating equipment and instruments
- Introducing machinery and instruments to facilitate goods storage
- Maintaining material handling equipment
- Maintaining logistics systems

The family **has 4 key job roles:** Maintenance director, Technical service manager, Technical service lead engineer, Technical service engineer.

Advanced Maintenance

The job family clusters roles that deal with **commissioning and continuous maintenance of cutting-edge technologies, machinery, and robotic systems** in logistics applications, ensuring their longevity through modern and specialized maintenance approaches.

Key tasks/activities include:

- Deploying autonomous picking robots
- Integrating advanced diagnostics tools
- Overseeing the installation of innovative maintenance solutions

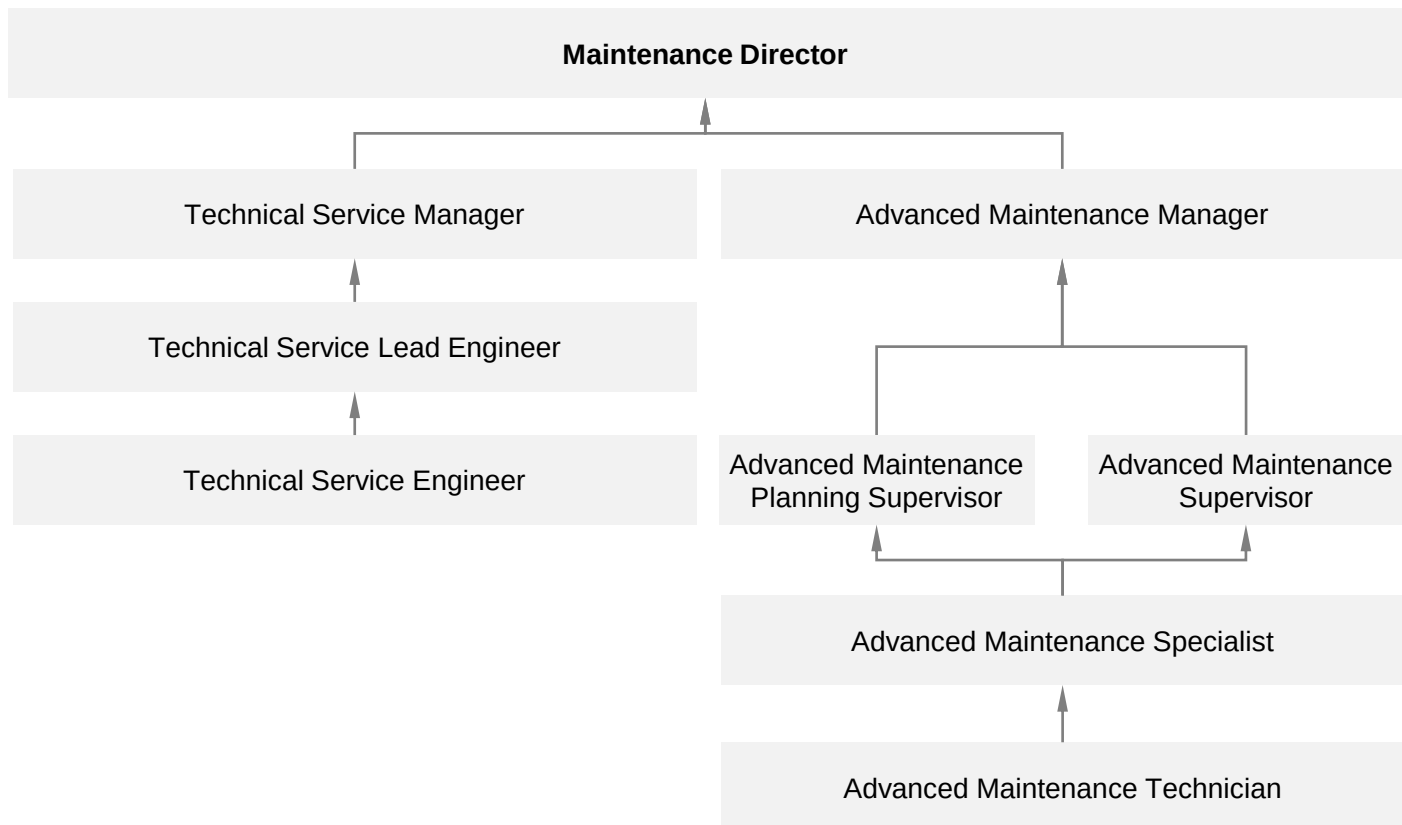
The family **has 6 key job roles:** Maintenance director, Advanced maintenance manager, Advanced maintenance planning supervisor, Advanced maintenance supervisor, Advanced maintenance specialist, Advanced maintenance technician.

Job family profiles and career pathway maps: (27) Engineering and Technical Service (28) Advanced Maintenance

Logistics equipment and systems maintenance (*related job families presented together*)

Engineering and Technical Service

Advanced Maintenance



Relevant USOC unit-tier codes for the job family: 1321 Manufacturing managers, 2144 Mechanical engineers, 2151 Electrical engineers, 1120 Managing directors and chief executives, 3113 Electrical engineering technicians, 3115 Mechanical engineering technicians.

Skills requirements

The job roles under Engineering and Technical services require skills related to equipment maintenance. The main skills include ground support equipment maintenance, design customization, machining, etc. While the job roles under Advanced Maintenance require skills related to systems and solutions maintenance. The main skills include automation design, autonomous systems engineering and maintenance, etc.

Full list of key skills is present in the following tables (see next page) providing a detailed view on job families' skills needs.

Job family profiles and career pathway maps:

(27) Engineering and Technical Service (28) Advanced Maintenance

Key relevant skills: Engineering and Technical Service

- Ground support equipment maintenance
- Design customization
- Emergency response management
- Engineering calculations
- Engineering maintenance management
- Engineering project management
- Engineering safety and security standards
- Equipment maintenance
- Machining
- Robotics and automation application

Key relevant skills: Advanced Maintenance

- Aircraft maintenance (MRO)
- Automation Design
- Autonomous systems engineering and maintenance
- Installation and assembly
- Material inspection
- Preventive maintenance
- Process analytical technology implementation
- Process improvement and optimization
- Safety and regulatory compliance
- Technology integration and application

Relevant certifications/ licenses (non-exhaustive)

Both the job families are rooted in strong engineering expertise with a **significant presence of mechanical and electrical engineers and technicians** for the roles. **Membership with the Saudi Council of Engineers** is mandatory for all practicing engineers and technicians in the Kingdom and the requirement is therefore also applicable to the engineering roles under these job families. In addition to this, there are various highly acknowledged certifications for these roles (mentioned below) that are prevalent in the Kingdom.

Certifications: Engineering and Technical Service

- Fundamentals of Engineering Certificate (FE) by National Council of Examiners for Engineering and Surveying
- Certified Six Sigma Black Belt

Certifications: Advanced Maintenance

- Fundamentals of Engineering Certificate (FE) by National Council of Examiners for Engineering and Surveying
- Certified Maintenance and Reliability Professional (CMRP) credential by Society for Maintenance and Reliability Professionals

Job family profiles and career pathway maps:

(29) Health, Safety, Security, Environment and Quality

(30) Regulatory Compliance

Job family overview

Health, Safety, Security, Environment and Quality and Regulatory Compliance job families have closely related job roles **contributing to safety, security and compliance in logistics operations**. Given the synergies, the **two families are presented on a single career pathway map**.

Health, Safety, Security, Environment and Quality

The job family clusters roles that ensure operations are conducted in a **safe, healthy, sustainable environment**, complying with national and international health and safety standards.

Key tasks/activities include:

- Implementing trade compliance solutions
- Developing, implementing, and monitoring HSE policies and procedures
- Conducting risk assessments
- Providing emergency services
- Conducting quality checks
- Providing training

The family **has 4 key job roles**:

General manager, Health, safety, security, environment and quality (HSSEQ)
manager, Health, safety, security, environment and quality (HSSEQ)
supervisor, Health, safety, security, environment and quality (HSSEQ)
analyst

Regulatory Compliance

The job family clusters roles that **focus on providing consulting services to facilitate compliance with international trade regulations and standards**, ensuring smooth cross-border transactions.

Key tasks/activities include:

- Implementation of trade compliance systems
- Conducting regulatory monitoring
- Providing expert advisory services to navigate complex trade environments
- Supporting risk assessments and audits
- Monitoring and assessing trade agreements

The family **has 4 key job roles**:

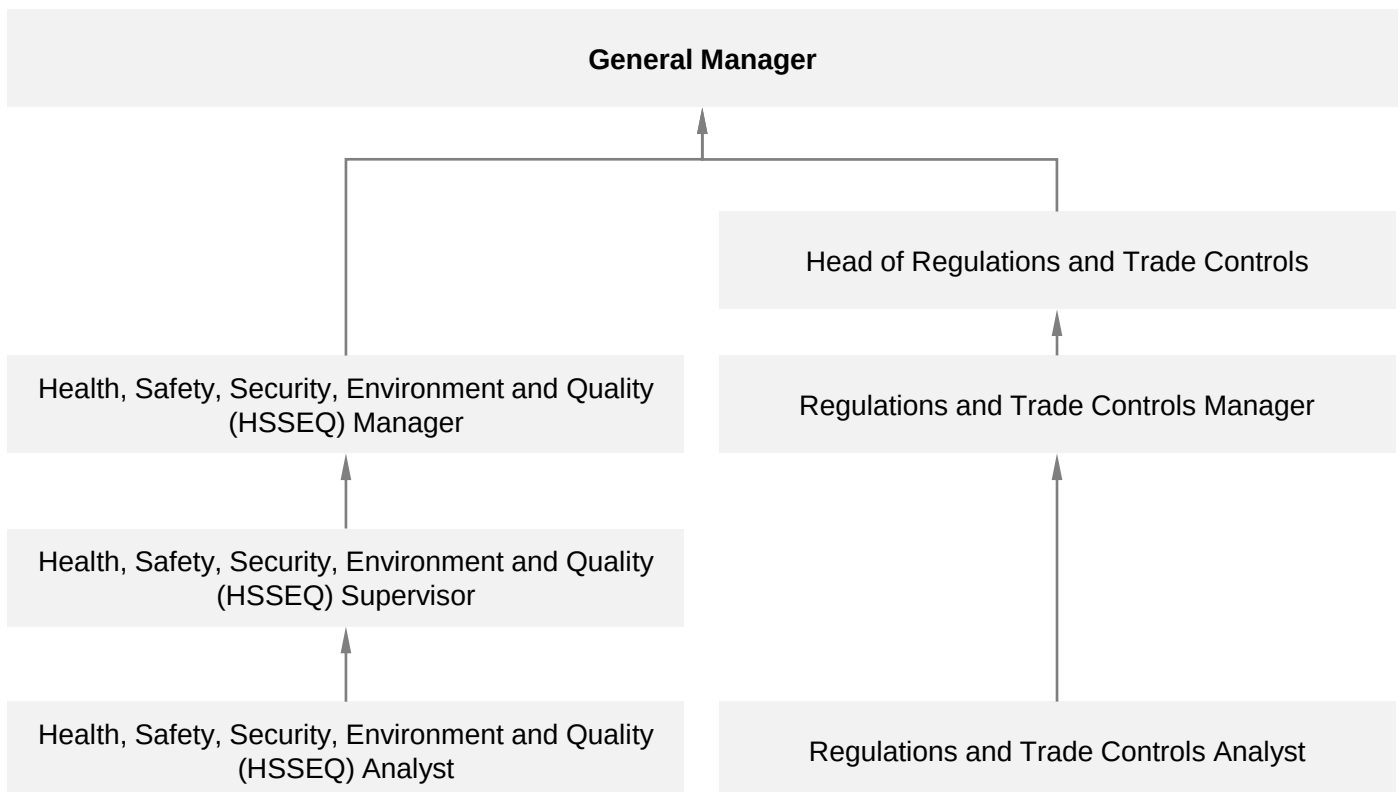
General manager, Head of regulations and trade controls, Regulations and trade controls manager, Regulations and trade controls analyst

Job family profiles and career pathway maps: (29) Health, Safety, Security, Environment and Quality (30) Regulatory Compliance

Safety and compliance (*related job families presented together*)

Health, Safety, Security, Environment and Quality

Regulatory Compliance



Relevant USOC unit-tier codes for the job family: 1213 Policy and planning managers, 1349 Professional services managers not elsewhere classified, 2263 Environmental and occupational health and hygiene professionals, 2421 Management and organization analysts, 3257 Environmental and occupational health inspectors and associates, 1120 Managing directors and chief executives, 1213 Policy and planning managers, 1349 Professional services managers not elsewhere classified, 2421 Management and organization analysts.

Job family profiles and career pathway maps:

(29) Health, Safety, Security, Environment and Quality

(30) Regulatory Compliance

Skills requirements

Health, safety, security, environment, and quality job family's roles require skills related to occupational hygiene, security, and quality. The main skills include defensive and economic driving techniques, hazardous materials identification system (HMIS) administration, etc. Regulatory compliance job family's roles are more focused on effective policy and regulations implementation. The main skills include corporate governance, data governance, dangerous goods management, etc.

Full list of key skills is present in the following tables (given below) providing a detailed view on job family skills needs.

Key relevant skills: Health, Safety, Security, Environment and Quality

- | | |
|---|--|
| • Defensive and economic driving techniques | • Warehousing/ cargo-related occupational health and safety management |
| • Hazardous materials identification system (HMIS) administration | • Warehouse facility management |
| • Risk Management and Administration | • Warehouse performance measurement |
| • Safety and regulatory compliance | • Workplace safety and health for vehicle operations |
| • Technology integration and application | • Workplace traffic safety management |

Key relevant skills: Regulatory Compliance

- | | |
|---|--|
| • Corporate governance | • Risk Management and Administration |
| • Data governance | • Regulatory knowledge |
| • Dangerous goods management | • Regulatory risk assessment |
| • Emergency response management | • Warehousing/ cargo-related occupational health and safety management |
| • Environmental sustainability management | |
| • Hazard and risk control and policy management | |

Job family profiles and career pathway maps:

(29) Health, Safety, Security, Environment and Quality

(30) Regulatory Compliance

Relevant certifications/ licenses (non-exhaustive)

The job family roles require strong expertise in **safety, quality assurance and compliance**. there are various highly acknowledged certifications for these roles (mentioned below) that are prevalent in the Kingdom.

Certifications: Health, Safety, Security, Environment and Quality

- NEBOSH International General Certificate on Health and Safety by National Examination Board in Occupational Safety and Health (NEBOSH), UK
- Certified Operational Safety Auditor by Board of Certified Safety Professionals (BCSP)
- Certified Operational Safety Auditor
- Automated Safety Systems Expert

Certifications: Regulatory Compliance

- Professional Certificate for Certified Compliance Officer (CCO)
- Global Trade Certification by Bankers Association for Finance and Trade (BAFT)

05

Appendix



Detailed view on skills needs emerging from sector transformations: Technology and digitalization

Transformation	Skill identified	Skill definition and relevance
Internet of Things (IoTs)	IoT applications	Implement Internet of Things (IoT) technologies to drive efficiency and effectiveness of operations
	Big data analytics	Analyze and validate significant volumes of data to discover and quantify patterns and trends to improve business operations
	Data governance	Develop, implement and practice guidelines, laws, and regulations across the organization for the handling of data at various stages in its lifecycle as well as the provision of advice on proper data handling and resolution of data breaches
Analytics and AI	AI applications	Facilitate the design and implementation of augmented reality applications to increase efficiency of work processes
	Data analytics	Organize and analyze structured or unstructured data to create insights systematically
	Cybersecurity	Understand, develop and apply cybersecurity policies and procedures to ensure protection against cybersecurity risks and vulnerabilities, and to respond to cybersecurity breaches
Blockchain	Blockchain technology application	Utilize distributed ledger technologies to automate business processes, improve quality assurance, traceability and supply chain transparency, and to facilitate secure collaboration and data sharing
	Technology integration	Integrate technologies into business operations of the organization to optimize efficiency and effectiveness of processes
	Information technology and network security	Manage organization's network and data security, ensuring an ongoing rigorous review of the organization's digital, cyber and application security and IT network infrastructures, to ensure multiple layers of defenses to protect proprietary data from attack and the organization's reputation

Detailed view on skills needs emerging from sector transformations: Technology and digitalization

Transformation	Skill identified	Skill definition and relevance
Autonomous transportation	Autonomous systems engineering and maintenance	Engineer, optimize, and maintain control systems for autonomous vehicles and drones to ensure efficient operation and sustainability with focus on predictive maintenance techniques
	Data analysis and machine learning in transportation	Employ data analytics and machine learning to interpret complex mobility data from various autonomous platforms, guiding user experience enhancements and smart maintenance schedules
	Automated Logistics and Port Infrastructure Management	Manage the implementation and upkeep of automated logistics systems, including the specialized equipment required for advanced port operations and autonomous transportation solutions
Advanced autonomous tools	Automated warehousing systems integration	Leverage advanced technologies and ERP systems like Oracle and SAP (Logistics modules) to automate and enhance warehouse operations, minimizing the need for manual input and maximizing efficiency and data accuracy
	Cobots in logistics	Enhance efficiency in warehouse processes through Human-Robot Collaboration (HRC), with cobots supporting activities such as order picking and inventory control
	Logistics risk and technology management	Address risks associated with sophisticated logistics environments, ensuring safety protocols keep pace with the demands of automated and cobotic systems in warehousing operations
High speed rails	Passenger train operations	Operate passenger train and respond to degraded and emergency train operations in accordance with regulatory and organizational requirements
	Railway safety and risk assessment	Expertise in railway safety regulations, risk assessment methodologies, and safety management systems to identify potential hazards, implement safety protocols
	Rail regulatory compliance	Adhere to rail legislative and regulatory requirements by establishing and developing internal work processes, to prevent incidents of non-compliance and regulatory oversight

Detailed view on skills needs emerging from sector transformations: Technology and digitalization

Transformation	Skill identified	Skill definition and relevance
Single Portal	Digital literacy	Mastering digital tools and platforms crucial for streamlining trade and transport operations through unified single-entry systems
	System connectivity	Establish and maintain seamless information sharing and data flow among various stakeholders and systems
	User experience design	Conceptualize and enhance the users' interactions and engagement with products and services by integrating elements of interaction design, information architecture, information design, visual interface design, user assistance design and user-centered design
3D printing technology	Rapid prototyping coordination	Overseeing the prototyping process, ensuring timely delivery of prototypes, and aligning design iterations with logistics requirements
	3D printing technical operation	Operating 3D printers, including assembly, calibration, monitoring print jobs, and performing regular maintenance for consistent output quality
	File preparation	Preparing and optimizing 3D print files, ensuring models are accurately rendered and print-ready, considering factors like material behavior during transport

Detailed view on skills needs emerging from sector transformations: Environmental, social and governance

Transformation	Skill identified	Skill definition and relevance
Environmental regulations and governance	Environmental protection management	Establish policies and procedures for sustainable environment practices covering green procurement, gas emissions, disposal methods, product quality standards and regulatory compliance
	Corporate governance	Establish guide and endorse organization's corporate governance and compliance policies. This includes being aware of the regulatory frameworks and global leading practices in similar organizations.
	Risk and compliance reporting	Develop regulatory documents required to meet regulatory reporting requirements and obligations
Supply chain sustainability	Environmental sustainability management	Integrate environmental sustainability through the development, implementation and review of sustainability strategies and programs against industry best practices
	Energy efficiency	Optimize efficient energy utility while in operation through the formulation and implementation of management plans
	Regulatory compliance	Develop, enforce, implement and practice regulatory compliance activities to ensure compliance to legislative and regulatory requirements
Circularity	Circular economy strategies	Process and energy strategies for a circular economy (incl. reverse logistics)
	Order fulfilment administration	Administer receiving, processing, delivery and optimization processes for orders to support business and customer requirements
	Regulatory compliance	Develop, enforce, implement and practice regulatory compliance activities to ensure compliance to legislative and regulatory requirements

Detailed view on skills needs emerging from sector transformations: Business ecosystem

Transformation	Skill identified	Skill definition and relevance
E-commerce	Order fulfilment administration	Administer receiving, processing, delivery and optimization processes for orders in order to support business and customer requirements
	Route planning and operations	Utilize data analysis to define the most efficient transportation paths and oversees the daily operational execution to optimize delivery schedules, reduce costs, and ensure prompt and precise order fulfillment
	Customer management	Manage customers across the customer lifecycle to guide customer interactions, with the goal of improving business relationships with customers, assisting in customer retention and driving sales growth
Supply chain robustness	Demand forecasting	Anticipate future demand for products or services by examining historical data and monitoring market trends, enabling strategic planning and allocation of the required resources, such as warehouse facilities and manpower
	Contingency planning	Develop alternate plans and protocols that can be activated as a response to unforeseen circumstances
	Transportation route optimization	Assess global political, economic, and social factors to identify efficient and secure transport routes, enabling proactive adjustments to supply chain operations and mitigating potential disruptions caused by geopolitical uncertainties
Partnerships	Third-party logistics provider selection and management	Select and manage third-party logistics providers to assess their capabilities, performance, and cost-effectiveness and optimize the supply chain
	Partnership management	Build cooperative partnerships with inter-organizational and external stakeholders and leveraging of relations to meet organizational objectives. This includes coordination and strategizing with internal and external stakeholders through close cooperation and exchange of information to solve problems
	Supply chain partnership risk management	Evaluate and mitigate disruptions that can affect partnerships in logistics, such as carrier delays or warehouse issues

Detailed view on skills needs emerging from sector transformations: Business ecosystem

Transformation	Skill identified	Skill definition and relevance
New business models	Last-mile delivery optimization	Enhance the final delivery phase to meet customer expectations for speed, efficiency, and reliability, especially in response to e-commerce growth
	On-demand logistics coordination	Coordinate real-time, flexible delivery networks that cater to on-demand and same-day delivery expectations of customers
	Logistics risk management	Proactive assessment and prioritization of logistics-specific risks associated with new business models, using targeted strategies to mitigate the impact on operations and customer service
Transport hub privatization	Transport hub investment analysis	Assess the viability and structuring public-private agreements for privatized transport hubs, with a focus on investment strategy and asset management
	Regulatory knowledge	Establish compliance by gaining a deep understanding of the regulatory frameworks governing transportation privatization policies
	Operational risk management	Manage failed procedures, system failures, policy errors and other business disruption events by assessing risks and implementing risk controls
Growth of cost-efficient carriers	Logistics solutions sales	Identify customer needs, evaluate these needs and identify probable solutions so as to sell logistics solutions and services to solve the needs of customers
	Value engineering	Apply value principles to increase productivity and efficiency in production lines by reducing operational costs and waste
	Transport cost optimization	Review and manage transportation costs, identifying savings opportunities across the logistics chain, and implement measures to enhance operational efficiency
Nearshoring	Cross Docking	Transfer inbound materials, goods and products from receiving docks to shipment docks for outbound deliveries
	Warehouse space utilization	Optimize spaces for storage of items in warehouses
	Warehouse inventory control/audit	Regulate outflows and inflows of stocked items as well as shipment, loading and storage activities using inventory control systems, methodologies and tools

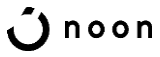
Detailed view on skills needs emerging from sector transformations: Customer preferences

Transformation	Skill identified	Skill definition and relevance
Transparency and visibility	Technology application	Apply and integrate evaluated technologies into organization operations or processes to achieve desired outcomes, including technologies allowing for higher transparency and visibility, e.g., Blockchain technologies, mobile applications
	Pricing for transportation services and operations	Apply knowledge in transportation services and operations to develop reliable and affordable transportation pricing models
	Customer experience management	Compile and analyze information gathered through various channels and manage communication across customer touch points to ensure a consistent and pleasant customer experience
Personalization	Market knowledge	Awareness and understanding of consumer demand, competitive landscape, and industry trends essential for logistics and transportation planning
	User experience design	Conceptualize and enhance the users' interactions and engagement with products and services by integrating elements of interaction design, information architecture, information design, visual interface design, user assistance design and user-centered design
	Strategic service excellence	Establish strategies and operating principles to consistently meet and manage customer's expectations in order to support business requirements
Omnichannel utilization	Analytical skills (incl. customer data analysis)	Devise frameworks for customer data analysis to develop an understanding of customer knowledge and behavior from various customer touchpoints
	Customer experience management	Compile and analyze information gathered through various channels and manage communication across customer touch points to ensure a consistent and pleasant customer experience
	Omnichannel supply continuity	Anticipate and manage risks to maintain an uninterrupted flow of goods and services across all transactional channels to ensure consistent access and fulfilment reliability
Micro-mobility	Technology Application	Integrate technologies into operations to optimize efficiency and effectiveness of processes
	Asset Management	Formulate and implement the organization's asset management policies to optimize asset life-cycle and performance
	Security management	Develop and enforce security measures in public transport premises in accordance with organizational standards and regulatory requirements

Detailed view on skills needs emerging from sector transformations: Customer preferences

Transformation	Skill identified	Skill definition and relevance
Mobility as a service (MaaS)	Market research and analysis	Proficiency in conducting market research and analysis to gain insights into customer behavior, trends, and preferences, enabling the development of strategies that aligns with evolving market needs
	User experience design	Conceptualize and enhance the users' interactions and engagement with products and services by integrating elements of interaction design, information architecture, information design, visual interface design, user assistance design and user-centered design
	Cybersecurity	Understand, develop and apply cybersecurity policies and procedures to ensure protection against cybersecurity risks and vulnerabilities, and to respond to cybersecurity breaches
Private aviation services	Flight planning	Plan cross-country flights according to regulatory and safety standards
	Flight dispatch	Review flight dispatch documents to identify dispatch issues and determine mission continuance
	Hazard and risk control and policy management	Apply systematic and objective approaches for hazard identification and risk assessment to manage workplace hazards effectively
Multimodality transportation	Logistics product solution	Manage activities to meet project requirements and translate to plans that deliver on time, on-budget learning and integration that organisations need
	Depot operations	Manage depot operations through activities for allocation and dispatch of mode of transport, Captain duty allocation, and facilities maintenance
	Interchange operations	Manage interchange operations through activities for service frequency regulation, readiness checks, safety management and crowd control, traffic control, customer service, and facilities maintenance

Sources

Category	Sources
Reports	Labour Activity in Transportation and Logistics Annual Report 
	The Future of Sustainable Transportation in Saudi Arabia 
	International Civil Aviation Organization reports 
	International Maritime Organization reports 
	EcommerceDB reports 
	Euromonitor International reports 
National strategies	Vision 2030 National Transformation Program 
	National Industrial Development and Logistics Program (NIDLP) 
	Human Capital Strategy 
	National Transport and Logistics Strategy 
	Saudi Aviation Strategy 
	PIF Key sectoral initiatives for Logistics Services and Transportation (2021-2025) 
Key players' strategies	Riyadh Metro Project Strategy 
	Jeddah Islamic Port Expansion Project 
	Launching of General Authority for Ports (Mawani) 
	Green Ports initiative, General Authority for Ports (Mawani) 
	DHL Supply Chain company strategy 
	Noon company strategy 
Benchmarks	Aramex company strategy 
	Skills Future  
	Training Packages  
	National Skill Development Corporation  
	HRDCorp. IndSF  
	Institute for Apprenticeships and Technical Education  
	European Skills, Competences, Qualifications and Occupations  

Glossary

Term	Definition
Artificial Intelligence (AI)	Artificial Intelligence, a field of computer science dedicated to creating systems capable of performing tasks that usually require human intelligence
Automated sorting systems	Technology-driven systems that use sensors, conveyors, and robotics to sort packages and goods automatically. These systems enhance efficiency and accuracy in parcel handling and distribution centers
Big data	The collection and analysis of large volumes of diverse and complex data sets to uncover patterns, trends, and insights
Blockchain	A distributed ledger technology providing a secure and transparent way to record transactions and track assets in a supply chain. It enhances traceability, reduces fraud, and improves the efficiency of logistics operations
Customer Experience (CX)	Refers to a customer's holistic perception of their experience with a business or brand
Digitalization	Digitalization is the use of digital technologies to change a business model and provide new revenue and value-producing opportunities
Freight transport	The process of transporting goods, commodities, and cargo by land, sea, or air. Efficient freight transport is critical for supply chain management and global trade, encompassing a range of vehicles and logistics services
Internet of Things (IoT)	The network of physical objects embedded with software, sensors, and other technology to connect and exchange data
Labor market	Labor market refers to the supply of and demand for labor, for which employees provide the supply and employers provide the demand
Labor productivity	Labor productivity, also known as workforce productivity, is defined as real economic output per labor hour
Logistics risk management	The identification, assessment, and mitigation of risks within the logistics and supply chain processes. It involves strategies to handle disruptions, ensure continuity, and safeguard assets and operations against uncertainties
Mobility as a Service (MaaS)	Mobility as a Service (MaaS) integrates multiple transport services into one platform, enabling seamless planning, booking, and payment across modes for a unified, on-demand transportation experience

Glossary

Term	Definition
Multimodality transportation	The use of multiple modes of transport within a single trip to optimize efficiency and convenience. This approach promotes the seamless integration of different transportation networks such as rail, road, and air
Nearshoring	The practice of relocating business processes or production closer to the home country to reduce transportation time and costs. Nearshoring enhances supply chain efficiency and reduces risks associated with long-distance shipping
Occupational safety and health	Promotion and maintenance of physical, mental and social well-being of workers in all occupations and prevention of workers from any departures of health caused by their working conditions
Private aviation services	The provision of individualized air travel services using private jets or chartered flights
Sector intelligence	Sector intelligence refers to analyzing a large group of companies operating similar businesses, to gain insights into economic trends and patterns
Sector Skills Council	Sector Skills Councils are employer-led national entities that undertake functions in skills training, education and workforce development
Skills gap	Skills gap refers to the disparity between the skills an employer expects their employees to have, and the actual skills employees possess
Sustainability	The maintenance and preservation of environmental, social, and economic resources to ensure long-term viability and welfare
The International Standard Industrial Classification (ISIC)	A United Nations industry classification system. Wide use has been made of ISIC in classifying data according to kind of economic activity in the fields of employment and health data

Abbreviations

List of abbreviations

AI	Artificial Intelligence
ASCM	Association for Supply Chain Management
CILT	Chartered Institute of Logistics and Transport
EV	Electric Vehicles
HRC	Human Robot Collaboration
IoT	Internet of Things
ISIC	International Standard Industrial Classification
MaaS	Mobility as a Service
MHRSD	Ministry of Human Resources and Social Development
MRO	Maintenance, Repair, and Overhaul
NIDLP	National Industrial Development and Logistics Program
SSC	Sector Skills Council
TMS	Transportation Management System
USOC	Unified Saudi Occupation Classification
WMS	Warehouse Management System

Acknowledgements

Key contributors

وزارة النقل والخدمات
اللوجستية
Ministry of Transport
and Logistic Services



- Badr Alghazi, Chairperson - Logistics Services and Transportation SSC) and Advisor on Human Capability Development
- Khalid AIRbaiyan, ADM for Logistics Services

Sector Expert

- Dr. Abdullah Alabdulkarim
- Adam Daoud, Operations Manager, Saudi Logistics Academy



- Dr. Mashael Algahtani, Director of Empowerment and Nationalization



- Mohammed Fallatah, GM Human Capital
- Saad Al Motairi, Regional OPS manager
- Orwa Sakatawi, HSE Manager
- Tamim Mohorgi, CRM and SLS Manager



- Rakan Alotaishan, Representative of Federation of Saudi Chamber for Logistics Services and Transportation SSC
- Khalid Albawardi



- Dr. Ibrahim Azmqna, Regional Operations Manager - GCC Countries
- Naif Alzahrani, Senior HR Business Partner



- Sara Aloraimi, SNC Representative for Logistics Services and Transportation SSC
- Abdullah Rashid Al Harbi, Human Resources Officer, NAQEL
- Salah Alabbad,
- Khoulod AlAhmari



- Khlaide Albusayyes, PIF Representative for Logistics Services and Transportation SSC
- Muath AlFraihi, Public Investment Fund Representative
- Bakr Almuhanah



- Eng. Abdulrahman Turkistani, GM - Human Resources and Business Support
- Othman bin Ibrahim Al-Sahil, HR Business Partner



- Dr. Omaimah Bamasag, DM - Transportation Enabling
- Osama Suliman Al Nuwiser, GM of Human Capital



- Saeed Alzahrani, Senior Director, Talent Acquisition and Organization Development
- Syed Ali - Advisor, Supply Chain & Logistics

Acknowledgements

Key contributors



- Sultan Al-Rasheedi, Executive Director of Business Development



- Hisham AlKhaldi, Chief Sector Officer



- Waleed Almalki, VP of Human Resources
- Ahmed Babatin, Former VP of Human Resources



- Saleh Al Daini, Chief Human Capital Officer



- Abdullah Al Ahmed, VP Development
- Sultan Alghamdi, General Manager



- Nada H. Al-Tamimi, Senior Director HR



- Samir Berqdar, CEO



- Fawaz Almuhanaa, Regional HR Manager



- Deena Faki, Area HR Manager



المجالس القطاعية للمهارات
Sector Skills Councils